

Unequal Job Opportunities and Well-Being Inequality: A Latent-Jobs Structural Decomposition

Hisham Haydar
University of Luxembourg and LISER

Build: 09 September 2026
Discussion draft

Abstract

We study how unequal job opportunities contribute to inequality in money-metric well-being. Our normative reference retains each household's own preferences and its own set of reachable jobs, while assigning the same disposable-consumption level to every job in that reference set; the money metric is the level at which the household's ex-ante evaluation of the reference matches the evaluation of the prospect it actually faces. We model labour supply as a choice among latent jobs and estimate preferences, job access and earning opportunities jointly on French EU-SILC data, with EUROMOD computing taxes, benefits and disposable income at every alternative work arrangement. We then introduce and implement a structural decomposition of the resulting inequality: well-being is recomputed under counterfactual equalizations of preferences, job access, earning opportunities, and household resources and needs, and the interactions among them are allocated with a grouped Shapley–Owen–Shorrocks rule. In the baseline Gini decomposition, labour-market opportunities account for 55.9 per cent of well-being inequality among single-adult households, of which job access alone carries 49.2 per cent, against 9.9 per cent for preferences and 34.2 per cent for household resources and needs. Among couples the balance differs: earning opportunities and resources and needs account for 35.7 and 51.2 per cent, while job access accounts for 8.2 and preferences for 4.8. Access exceeds earning opportunities for single adults, and the ordering reverses for couples, under all six inequality indices we report; the sign of the single-adult preference contribution is not robust across indices.

Preliminary results. Intervals on the headline shares are cluster-robust parameter percentiles from 100 draws and are reported separately from the integration band, never merged with it. The subdivision of couples resources against household composition requires a repricing that has not been run and is reported jointly.

1 Introduction

Two people can work the same hours for the same hourly pay and be very differently placed. One chose that job from several that were open; the other took the only offer available. Their earnings are identical and their circumstances are not. Income comparisons record the outcome and cannot separate the two cases, and the difference matters for how unequal we should judge well-being to be.

The same ambiguity runs through the whole distribution. Low earnings can describe someone who values leisure highly and chose short hours, someone who cannot obtain a well-paid job, or someone whose family resources make a different work choice affordable. These are observationally entangled and economically different. A distribution of income is a distribution of outcomes with the explanations already mixed in.

The ambiguity matters for measurement, not only for interpretation. If we want to know how much of the inequality we observe is associated with unequal access to employment, hours

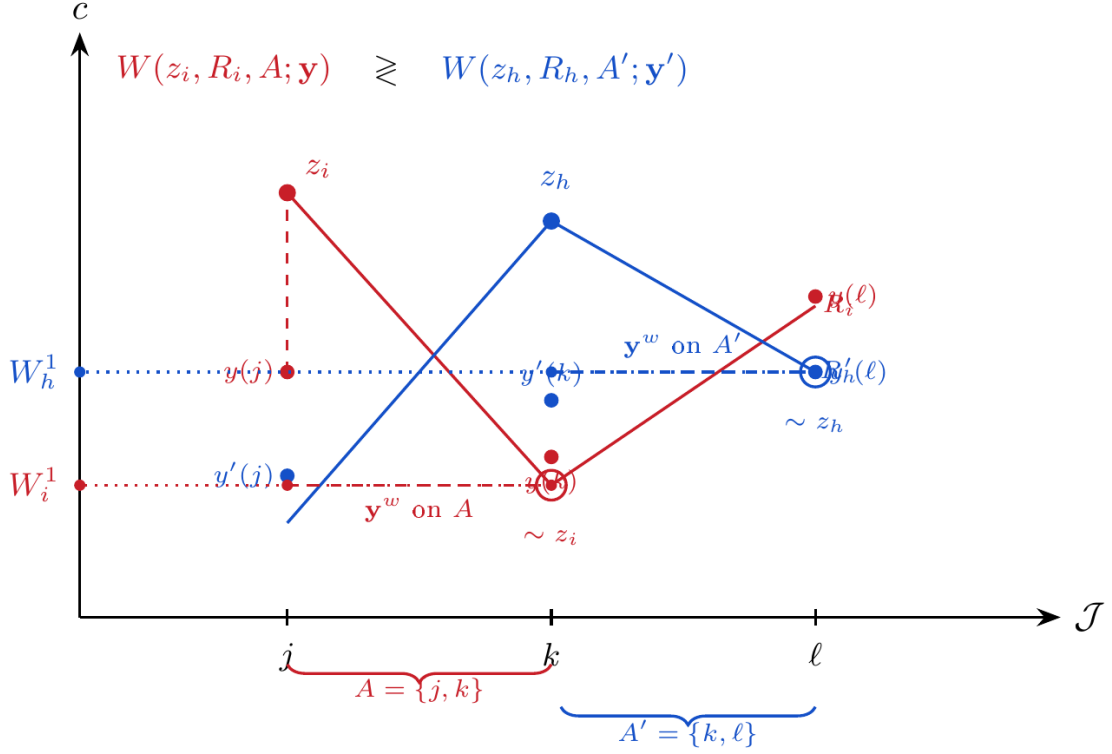


Figure 1: Own-set equal-consumption equivalents. Individuals with preferences R_i, R_h and ability sets $A = \{j, k\}$ and $A' = \{k, \ell\}$ attain z_i and z_h . For each individual, a common consumption level is assigned to every job in their own set; the level at which the preferred reference job becomes indifferent to the attained bundle is the money metric. Adapted from Haydar and Maniquet (2026), work in progress. This is the deterministic construction; the estimated measure is its ex-ante extension, defined in Section 4.

arrangements and wage offers, we need two things that an income distribution does not supply. We need a behavioural model that separates what a household wanted from what it could reach, and we need a rule for comparing people whose opportunities and whose tastes both differ. Neither is optional: without the first there is nothing to attribute, and without the second there is no defensible way to say who is better off.

The normative reference. Comparing well-being when individuals face different job opportunities requires a reference that specifies how those opportunities enter the comparison. We draw on the own-set equal-consumption criterion of Haydar and Maniquet (2026), work in progress. In its deterministic formulation, the criterion assigns to an attained situation the consumption level that would make the individual's preferred job within their own ability set equally good, when every feasible job provides that same consumption. The reference therefore retains the individual's own opportunities and own preferences while removing variation in pay from the reference bundles. Writing A_i for the jobs available to i and z_i for the attained situation, the reference level solves

$$u_i(z_i) = \max_{j \in A_i} u_i(W_i, j).$$

The argument being solved for is an amount of consumption, so the measure is money-metric by construction; there is no second conversion from an index into euros.

We adapt this principle to an estimated distribution of latent jobs. The empirical object is not the deterministic maximum but its ex-ante counterpart: the household's evaluation of the reference prospect, integrated over the jobs it may reach, is set equal to its evaluation of the

prospect it actually faces. Section 4 states the stochastic formulation, derives its closed form, and says precisely which properties of the deterministic criterion carry over and which are not claimed.

One comparative static is worth stating plainly at the outset, because it is easy to misread. Holding the attained situation fixed, a person with a larger reference menu may need less uniform consumption to reach the same satisfaction. When actual opportunities change, however, both the attainment and the reference menu change. The net effect on equivalent consumption depends on both. Measuring attainment against one’s own opportunities is therefore not a claim that larger menus are intrinsically better, and we make no such claim.

The empirical approach. The behavioural half is a random-utility, random-opportunity model of job choice. A job is a package: an employment state, an occupation, a weekly hours arrangement and an hourly wage. Households rank packages by consumption and leisure, and the packages differ in how available they are. Both the preferences and the household-specific intensity of availability enter one likelihood and are estimated jointly; neither is observed as a complete schedule. Every package is priced through the French tax-benefit system, so a change of hours or occupation moves disposable income through the actual schedule of taxes and transfers rather than a linear approximation. We estimate the model separately on 1,540 single-adult and 2,223 couple households drawn from French EU-SILC, with couples choosing jointly under a shared household budget.

The normative half then yields, for each household, an equivalent flat consumption level. To decompose its inequality we do not linearise. We define four structural equalization operators, one for preferences, one for job access, one for earning opportunities and one for household resources and needs; we recompute every household’s welfare level, and the inequality of the resulting distribution, under each of the sixteen coalitions of those operators; and we allocate the interactions among them with the grouped Shapley–Owen–Shorrocks rule. The allocation is exhaustive by construction and the closure is verified numerically rather than imposed.

What we find. Among single-adult households, labour-market opportunities carry 55.92 per cent of the inequality in the money metric, of which job access alone carries 49.16 per cent and earning opportunities 6.76. Preferences carry 9.88 per cent and household resources and needs 34.20. Among couples the composition is different rather than the total: earning opportunities carry 35.72 per cent and resources and needs 51.20, while job access carries only 8.23 and preferences 4.85. Single adults are access-dominated on the labour-market side and couples are earnings- and resource-dominated. We report that contrast as a finding and do not attach a mechanism to it.

Two qualifications belong with the headline. First, the ordering of access against earning opportunities is the robust part: access exceeds earning opportunities for single adults under all six inequality indices we report, and the ordering reverses for couples under all six. The complete ordering of all four components is not robust in that sense, and the sign of the single-adult preference contribution changes outside the Gini. Second, an allocated share is an average of marginal contributions over coalition orders. It is not the reduction that equalizing that group alone would achieve. For single adults the two differ instructively: equalizing preferences alone would *raise* the Gini by 13.0 per cent, while the allocated preference share is a positive 9.88 per cent. Section 5 explains why both statements are correct.

Relation to existing work. Every ingredient here has antecedents, and the contribution is the combination and the empirical answer, not any one step.

Modelling labour supply as choice among latent jobs, with preferences and offer intensities identified jointly from one likelihood, is due to [Aaberge et al. \(1995\)](#) and [Aaberge et al. \(1999\)](#), developed by [Dagsvik and Strøm \(2006\)](#) and [Dagsvik and Jia \(2016\)](#), surveyed by [Aaberge and Colombino \(2018\)](#), and set out for applied work by [Capéau et al. \(2016\)](#). We inherit that framework, including its distinction between the intensity of an offer and the probability of a choice, and its reliance on excluded opportunity shifters to separate components that choices

alone identify only jointly. [Capéau et al. \(2016\)](#) also supply two of the identifying restrictions we maintain: the wage-offer distribution is independent of offered hours, and a local unemployment measure shifts availability while being excluded from preferences. Their Belgian application already covers single women, single men and couples, so covering both household types is not itself a contribution.

Combining such a model with a monetary welfare evaluation is also established. [Aaberge et al. \(1995\)](#) translate attained utility into an equivalent income against a stated reference household, choice set and tax system, and [Aaberge et al. \(2004\)](#) evaluate reforms with joint household choice and restricted hours. [Jia and Thoresen \(2021\)](#) make the case for the job-choice model in exactly this role and report a distribution of compensating variation for a Norwegian reform, and [Jacquet et al. \(2026\)](#) compare a standard compensating variation with one computed after replacing preference characteristics by reference values. All four evaluate a *change* between policy regimes. Our object is a cross-sectional distribution of levels under one policy system, and its attribution. A reform gain and a level have different origins and different denominators; a small preference-related reform difference would not imply a small preference share in our decomposition, and our shares do not estimate their reform gains.

The normative half draws on the literature that refuses to resolve interpersonal comparison by assuming common preferences. [Fleurbaey and Maniquet \(2006\)](#), [Fleurbaey and Maniquet \(2017\)](#) and [Fleurbaey and Maniquet \(2018\)](#) set out how a reference bundle encodes a position on compensation and responsibility; [Decoster and Haan \(2015\)](#) and [Bargain et al. \(2013\)](#) show empirically how much a welfare ordering moves with that choice. Their references fix a wage or an unearned-income intercept and maximize over a deterministic budget set. Ours fixes consumption across the alternatives of an estimated opportunity distribution and integrates. That is a different reference and, as Section 6 records, the ordering is sensitive to it.

The closest decomposition precedents are two. [Mühlhan \(2023\)](#) combines a structural labour-supply model with involuntary-unemployment restrictions and a Shapley attribution, decomposing the change in German household *income* inequality into temporal factors and binary restrictions. [Creedy and Héroult \(2011\)](#), in the section that constructs money-metric distributions under alternative policy and population states, decompose a change in inequality and social welfare by averaging over the two orders in which policy and population can be changed; theirs is the closest money-metric welfare-inequality decomposition we know. Both are decompositions of a *change* between two situations, with policy, population or temporal factors as the factors. Ours is a decomposition of a cross-sectional *level* of well-being inequality, with the factors defined as structural operators inside an estimated job-choice model: what a household prefers, which jobs it can reach, what those jobs pay, and what its budget and needs are. The wider behavioural-simulation decomposition tradition, [Bargain \(2012\)](#) and [Jessen \(2019\)](#), decomposes income changes in the same spirit.

The allocation rule is inherited outright. [Shorrocks \(1982\)](#) states the accounting requirements a decomposition should satisfy; [Shorrocks \(2013\)](#) places the Shapley value at the centre of distributional decomposition; [Owen \(1977\)](#) supplies the value for games with a priori unions, which is what a grouped allocation requires; [Sastre and Trannoy \(2002\)](#) documents how much the answer can depend on how the exercise is set up; and [Audoly et al. \(2025\)](#) give the contemporary practitioner’s account of the grouped nonlinear case we use. We claim no new allocation principle. Complete recomputation of every coalition, rather than a linear approximation, likewise has precedents and is better described as methodological discipline than as a contribution.

What is new, to our knowledge, is the conjunction: a random-utility random-opportunity model of job choice, an opportunity-sensitive money-metric welfare level built on the own-set equal-consumption reference, a structural preference/access/earnings/resources game whose operators are defined inside that estimated model, complete recomputation of welfare and inequality under every coalition, and a grouped allocation of the result. The application-specific methodological contribution is the game and its counterfactual operators. The cooperative-game

allocation rule is inherited.

Roadmap. Section 2 describes the data and the household budget construction. Section 3 presents the latent-jobs model, its identifying restrictions and its estimation. Section 4 defines the money-metric welfare measure and the structural decomposition. Section 5 reports the behavioural and welfare results for both household types. Section 6 examines sensitivity and states the limitations. Section 7 concludes.

2 Data

The data are French EU-SILC, collection year 2016, with income reference year 2015. Taxes, benefits and disposable income are computed by EUROMOD ([Sutherland and Figari, 2013](#)) under the French 2015 policy system, which is the system in force over the income reference period. We use the same three dates consistently: the survey is collected in 2016, incomes refer to 2015, and the simulated policy system is 2015.

2.1 The estimation samples

Table 1: Sample construction. Unweighted households remaining after each screen, from the France 2016 EUROMOD input file to the two estimation samples. Rows are sequential; the counts are read from the frame records and are not reconstructed by subtraction.

Screen	Single-adult	Couple
the raw France 2016 input file	4,038	5,965
one- or two-adult households (composition screen)	4,038	5,965
every adult aged 20 to 60	2,131	3,662
no adult still in education	2,036	3,521
no old-age, disability or survivor pension receipt	1,755	3,218
labour status in scope	1,598	2,412
no other employable or earning member	1,564	2,323
observed hours and wage inside the calibrated support	1,555	2,275
hours outside [5,70] or unsupported military occupation (ISCO 0) on an employed decider	1,543	2,223
observed chosen alternative priced at non-positive disposable consumption	1,540	2,223
Estimation sample	1,540	2,223

The screens are of three kinds and it is worth separating them. The first two define the decision unit: the household must contain either one unpartnered adult or two mutually linked adults of opposite sex, and every decider must be between twenty and sixty. This is the largest exclusion and it is structural. Multi-generational households, flat-shares, adult children living with parents and same-sex couples are outside the estimated population, and no result extends to them. Because the age screen binds on both spouses, it removes proportionally more couples than singles.

The second kind removes households for whom the model does not define an offer set: adults in full-time education, households receiving an old-age, disability or survivor pension, and deciders whose labour-market status lies outside employment, unemployment and inactivity. Together with the pension screen this is why the surviving employment rate is high. The employment share below should be read as a share among people for whom working is a live option, not as a French employment rate.

The third kind is support. An employed decider’s observed hours must lie in the closed interval [5, 70] hours per week and the delivered hourly wage in [2, 590] euros per hour, because those are the boundaries

of the supports the opportunity densities are defined on. An observed occupation must map into the four modelled groups; ISCO 0, the armed forces, does not, and the resulting exclusion is *unsupported* occupation, not missing occupation. Three single-adult households whose own observed job prices to non-positive disposable consumption are removed, because the log-consumption term is undefined at their observed choice; non-positive *simulated* alternatives are not a reason to drop a household, and are instead excluded from that household’s choice domain.

Table 2: The four occupation groups. ISCO-08 major groups are aggregated into four modelled groups; group 1 is the omitted reference in the estimated occupation block. This is a research aggregation adopted for this paper, not an ILO classification. ISCO 0, the armed forces, has no modelled alternative and is a sample screen; that is unsupported occupation, not missing occupation.

Model group	ISCO-08 major groups	Description
1 (reference)	6–9	Skilled agricultural, craft, plant and machine operators, and elementary occupations
2	5	Service and sales workers
3	4	Clerical support workers
4	1–3	Managers, professionals, technicians and associate professionals

2.2 What the households look like

Table 3: Descriptive means on the estimation samples. Weighted by the household cross-sectional weight; one row per household, with spouse-specific variables carried on the household row. Hours and wages are conditional on employment. These are observed inputs, not model predictions.

Measure	Single-adult decider	Couple man	Couple woman
Age (years)	41.1	39.8	37.8
Children under 20	–	–	–
Usual weekly hours, employed	37.6	41.1	35.8
Delivered hourly wage, employed (EUR/hour)	–	–	–
Households (unweighted)	1,540	2,223	

Education is the three-level ISCED grouping, with the medium level as the omitted reference in the wage equation. Potential experience is years since leaving education, entering the wage equation in units of 20 years, with the square recomputed after scaling rather than carried forward. A child is any resident household member under 20, counted without a parent link. Age enters preferences centred on the sample mean decider age and divided by 10 years.

Table 4: Observed joint participation regimes among the 2,223 estimated couple households. These are joint regimes, not two spouse marginals.

Regime	Meaning	Weighted share	Mean hours, man	Mean hours, woman
BB	both spouses employed	0.8435	41.0	35.9
MO	man employed, woman not employed	0.0796	41.5	0.0
WO	woman employed, man not employed	0.0547	0.0	35.0

Regime	Meaning	Weighted share	Mean hours, man	Mean hours, woman
NN	neither spouse employed	0.0222	0.0	0.0

2.3 The household budget

For a given work arrangement the tax-benefit model is given the implied gross labour inputs, weekly hours, hourly wage, and earnings split at the thirty-five hour threshold, together with the household's non-labour inputs and roster, and returns disposable income. Only deciders receive counterfactual overrides; other members keep their baseline values. The accounting identity linking original income, benefits, taxes and social contributions to disposable income holds to machine precision at both the person and the household-alternative level, before and after the benefit take-up adjustment. Disposable income is summed over all resident members of the household.

Panel (c) of the figure reports the *incidence* of non-labour budget inputs by family, not a cash total. Some of these are stocks, some are annual flows and some are monthly; they are inputs to the tax-benefit calculation and are not added together into one income concept. The pension family is empty by construction, because pension receipt is a sample screen. Panel (a) reports the tax-benefit *output* at the observed job, which is an object of a different kind and is never added to panel (c).

Local labour-market conditions enter through a group unemployment measure, a population-weighted exposure index for the decider's region and age-education cell, stored as a fraction and entering the employment index multiplied by 10. Regional and urbanisation variation supports the access specification conditional on the exclusion restriction and the functional form; it does not by itself identify access, and no causal effect of geography is claimed anywhere in this paper.

3 A latent-jobs model of household labour supply

A job is a package j : an employment state $e \in \{0, 1\}$ and, when employed, an occupation $k \in \{1, \dots, 4\}$, weekly hours $h \in [5, 70]$ and an hourly wage $w \in [2, 590]$. The base measure ν places counting mass on the single non-employment point and, on the employed part, counting measure over occupations with Lebesgue measure over hours and wages. A household chooses one package; a couple chooses one pair of packages under a shared budget.

3.1 Preferences

Let $C_i(j)$ be priced monthly disposable consumption at j and $\ell_i(j)$ normalized leisure, $(80 - h)/10$ hours per week, floored at one hour. For a single adult of sex group g , deterministic utility is

$$u_i(j) = \underbrace{\omega_{ig} \text{BC}(\ell_i(j); \theta_{\ell g})}_{L_i(j)} + \beta_c \log(C_i(j)/\lambda_c), \quad \omega_{ig} = \beta_{\ell 0, g} + \beta_{\ell a, g} a_i + \beta_{\ell a^2, g} a_i^2 + \mathbf{1}\{g = f\} \beta_{\ell n, g} n_i,$$

where $\text{BC}(x; \theta) = (x^\theta - 1)/\theta$ and $\text{BC}(x; 0) = \log x$, a_i is centred age in decades, and n_i is the child count. For a couple the index is additive across spouses with no direct cross-leisure term,

$$u_i(j) = \omega_{im} \text{BC}(\ell_{im}; \theta_{\ell m}) + \omega_{if} \text{BC}(\ell_{if}; \theta_{\ell f}) + \beta_c \log(C_i(j)/\lambda_c),$$

with consumption the tax-unit sum over all members. We write $L_i(j)$ throughout for the complete non-consumption part of the index, which for a couple contains both leisure terms.

Three features of this specification are decisions and are worth naming. The consumption curvature is exactly zero, so the consumption term is $\beta_c \log(C/\lambda_c)$ and the marginal utility of consumption is β_c/C , independent of the normalizer. The consumption weight β_c is **estimated**, not fixed at one. And the random-utility shock is i.i.d. type-I extreme value with scale one, so utility is measured in the natural unit of that scale. Fixing the shock scale is a normalization; additionally fixing β_c would be a substantive restriction relative to that normalization, and estimating β_c removes it. Neither step identifies an absolute cardinal utility scale, and fixing the consumption functional form is not by itself what identifies the shock scale.

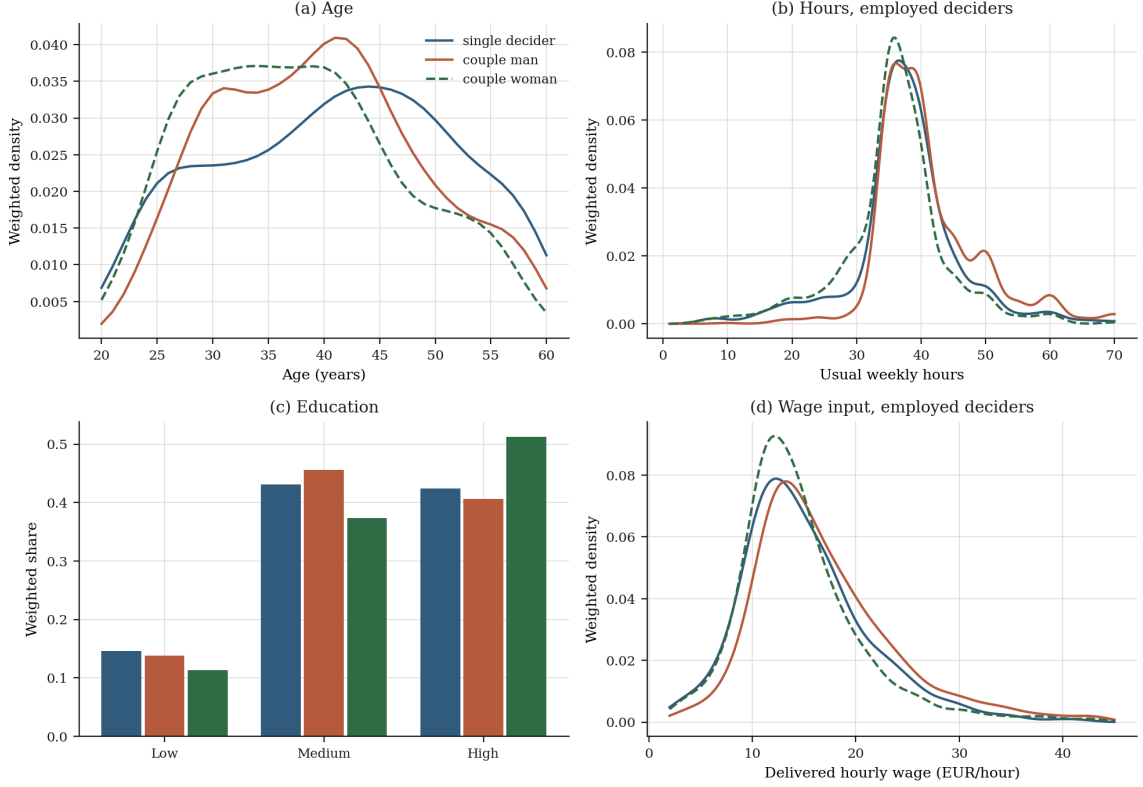


Figure 2: Weighted distributions on the estimation samples: age, usual weekly hours among employed deciders, education, and the delivered hourly wage input. Observed inputs, not model predictions.

The normalizer λ_c is a units convention. Under exact log consumption it enters the index as the alternative-invariant constant $-\beta_c \log \lambda_c$, so it cancels from every choice probability and, as shown below, exactly from the welfare measure.

3.2 Opportunities

The opportunity kernel gives the relative intensity with which packages are available to household i , before any choice is made. Up to a household constant it factors as

$$\log \tilde{g}_i(1, k, h, w) = \beta_E + x_i^{E'} \gamma + \xi_{k,g} + H(h) + \log \phi(w \mid \mu_i(k), \sigma), \quad \log \tilde{g}_i(0) = 0,$$

with four blocks. **Employment access** $\beta_E + x_i^{E'} \gamma$ carries the group unemployment measure, region and urbanisation, and multiplies the working indicator. **Occupation access** is $\xi_{k,g}$ with group one the omitted reference. The **hours density** $H(h)$ elevates five bands over a residual reference set of total width 26.5 hours per week; the narrow full-time band $[33.5, 36.5)$ is a density elevation over an interval, not an atom at thirty-five hours. The **wage density** ϕ is log-normal with location $\mu_i(k) = \beta_{w0} + \beta_{wL} \text{educ} L_i + \beta_{wH} \text{educ} H_i + \beta_{wx} x_i + \beta_{wx^2} x_i^2 + \delta_k$ and dispersion σ , truncated to $[2, 590]$ euros per hour and renormalized on that support.

Two properties of the normalization matter later. Write $\hat{g}_i = \tilde{g}_i / Z_i$ for the kernel normalized to a probability density on the whole package space. The wage factor integrates to one *conditional on employment and occupation*, because the truncated density is renormalized on its own support; and the non-consumption index L_i has no wage argument. Together these give the reference the pay-neutrality property established in Section 4.

The four blocks are what the decomposition later separates. Employment and occupation access are one mechanism; the distribution of wages conditional on an occupation is another; neither is the same as non-labour resources or household needs.

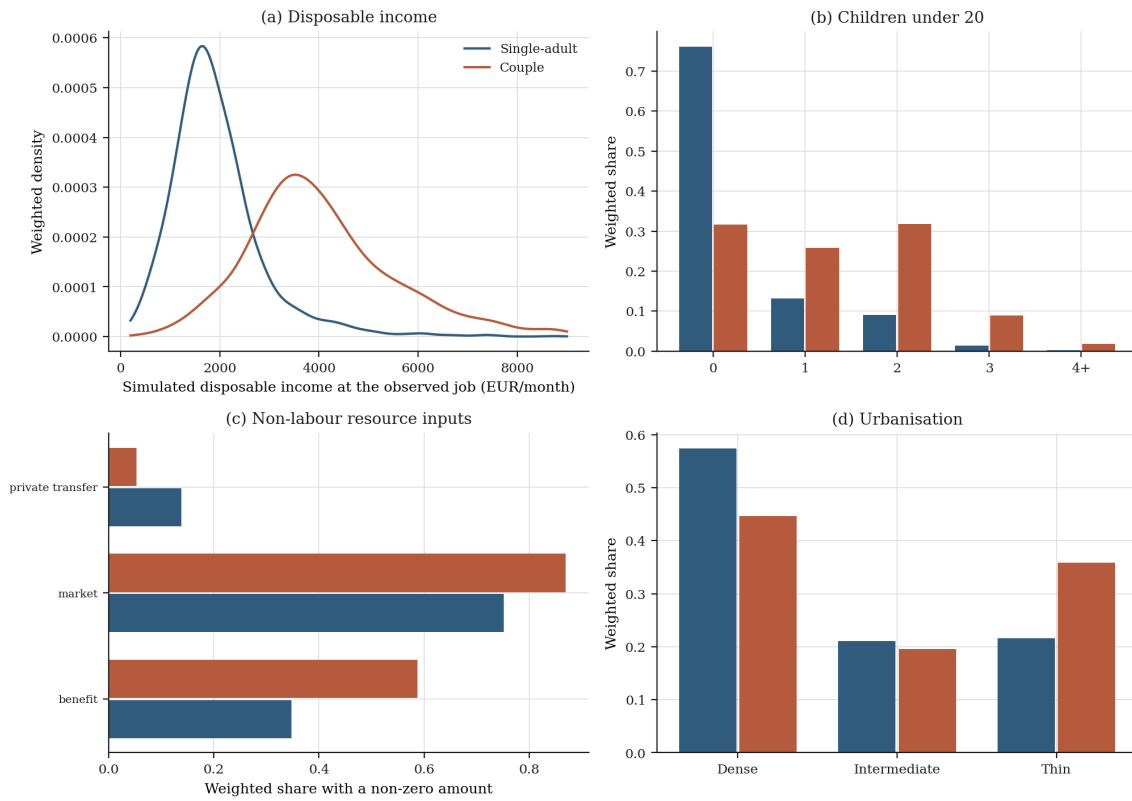


Figure 3: Weighted distributions on the estimation samples: simulated disposable income at the observed job, children under twenty, the incidence of non-labour budget inputs by family, and urbanisation. Panel (a) is a tax-benefit output evaluated at the observed choice; panel (c) reports inputs to that calculation. The two are never added, and stocks are never summed with monthly flows.

3.3 Identification, in words

Choices alone do not separate a taste for leisure from a scarcity of jobs at that number of hours. Three kinds of restriction do the work. Preferences are smooth in hours through the leisure index, while the opportunity density is a step function over bands, so a spike in observed hours at a band is read as availability rather than as a kink in tastes. Excluded shifters enter availability and not preferences: the local group unemployment measure and the regional and urbanisation indicators shift the employment index and appear nowhere in utility. And the wage-offer distribution is specified as independent of offered hours conditional on occupation, which separates the wage location from the hours density. These are the restrictions of [Capéau et al. \(2016\)](#) and [Dagsvik and Jia \(2016\)](#), maintained here rather than tested. The regional variation supports the access block conditional on those restrictions; it does not establish separate identification on its own, and it is not a causal design.

3.4 Estimation

The choice set is a continuum, so the likelihood is evaluated over sampled alternatives. For each household we draw 100 alternatives from a proposal density q_i and place the observed choice in the set as well, giving 101 rows. The contribution of household i is the conditional probability of its observed package within that set, corrected for the proposal:

$$\Pr(y_i \mid \text{set}) = \frac{n_y \exp\{a_i(y) - \log q_i(y)\}}{\sum_{s \in \text{slots}} \exp\{a_i(s) - \log q_i(s)\}}, \quad a_i(j) = u_i(j) + \log \tilde{g}_i(j).$$

The denominator runs over slots rather than distinct packages, and a package drawn more than once keeps its multiplicity, n_y ; the chosen package carries the same proposal correction as any other row. The proposal is fitted out of fold, so a household's own outcome never enters the proposal used to score it. For couples the proposal draws the joint participation regime first and then the two spouse packages conditional on it.

Standard errors are cluster-robust on the household. Optimization uses five starts under two polishing contracts, ten terminal paths in all; we report the spread of the criterion across those paths and the eigenvalues of the exact Hessian at the selected optimum. Those diagnostics support a stable local solution found from the starts tested. They are not a proof of global uniqueness, and we do not claim one.

Two features of the observation rule remain genuinely open and are stated here rather than in a footnote. The sample screens on the observed hours and wage of employed deciders, so the estimation sample is selected on an outcome of the process being modelled. The conditional likelihood above is the correct object for a sampled choice set; it is not automatically the correct object for an outcome-selected sample, and we have not established which correction, if any, the screen requires. Separately, the derivation of the sampled-set probability conditions on a labelled collection of slots; small observed cross-coordinate correlations and a binomial-looking multiplicity distribution are consistent with independent draws but do not by themselves establish the sampling law. Both are recorded as unresolved in Section 6.

4 Money-metric well-being and structural inequality decomposition

4.1 The measure

Fix a household i and a coalition state S , which determines the utility index $L_{i,S}$, the normalized opportunity density $\hat{g}_{i,S}$ and the priced consumption $C_{i,S}$ at every package. Define the attained ex-ante value and the reference mass

$$J_{i,S} = \int e^{L_{i,S}(j)} \left(\frac{C_{i,S}(j)}{\lambda_c} \right)^{\beta_c} \hat{g}_{i,S}(j) d\nu(j), \quad H_{i,S} = \int e^{L_{i,S}(j)} \hat{g}_{i,S}(j) d\nu(j).$$

The reference offers the same flat monthly amount m at every package while retaining that state's L and $\hat{g}$, so the value of the reference is

$$\Phi_{i,S}(m) = \beta_c \log(m/\lambda_c) + \log H_{i,S}, \quad \Phi'_{i,S}(m) = \beta_c/m > 0.$$

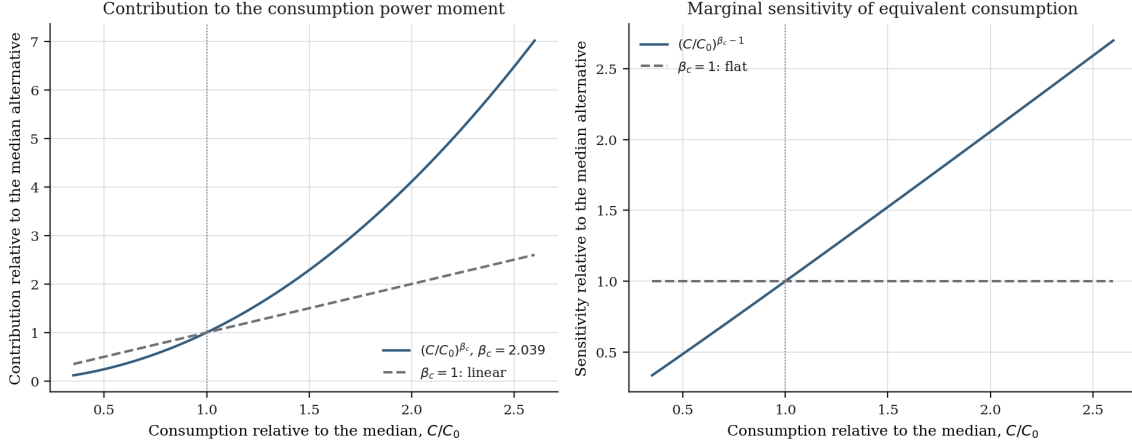


Figure 4: The consumption coefficient as the order of a power mean. Left: the contribution an alternative makes to the consumption power moment, relative to the median alternative, against the linear comparison at $\beta_c = 1$. Right: the marginal effect of that alternative's consumption on the resulting equivalent amount, which carries exponent $\beta_c - 1$, against the flat comparison at $\beta_c = 1$. Neither curve is a reference probability weight: holding the reference measure fixed, those weights do not vary with consumption at all.

The money metric is the amount at which the reference reaches the attained value, $\Phi_{i,S}(W_{i,S}) = \log J_{i,S}$, which inverts in closed form:

$$W_{i,S} = \lambda_c \exp \left[\frac{\log J_{i,S} - \log H_{i,S}}{\beta_c} \right].$$

Writing $r_{i,S}(dj) = e^{L_{i,S}(j)} \hat{g}_{i,S}(j) d\nu(j) / H_{i,S}$ for the reference probability measure, the same object is

$$W_{i,S} = \left[\int C_{i,S}(j)^{\beta_c} r_{i,S}(dj) \right]^{1/\beta_c}.$$

The measure is therefore a **weighted power mean of consumption of order β_c** , taken under a reference measure that weights packages by their non-consumption value and their availability. It is an arithmetic mean only at $\beta_c = 1$. The normalizer λ_c cancels between the two forms, which is why the constant is a units convention and not a modelling choice; we verify this numerically to a maximum relative deviation of 2.7e-15 across four widely separated values.

Three quantities are easy to conflate and are distinct. The reference probability weight r_j does not vary with consumption at all when the reference measure is held fixed. The *contribution* an alternative makes to the power moment is proportional to $C_j^{\beta_c}$, so an alternative paying twice the median contributes 4.11 times as much at the estimated single-adult coefficient. The *marginal effect* of that alternative's consumption on the resulting amount is $\partial W / \partial C_j = r_j C_j^{\beta_c - 1} W^{1 - \beta_c}$, with exponent $\beta_c - 1$. The figure plots the second and the third and labels each; neither is a statement that the alternative is more available.

β_c is the order of this power mean and the coefficient on log consumption. It is not the consumption curvature, which is exactly zero here, and it is not an inequality-aversion parameter across households, which belongs to the index applied in Section 5 and not to the household's own aggregator.

4.2 What the reference does and does not do to pay

Under the stated factorization the reference is invariant to the conditional wage density: that density integrates to one on its support and the non-consumption index has no wage argument, so replacing it leaves H_i unchanged. Numerically, the largest household change in $\log H$ is 1.8e-15 for single adults and 7.1e-15 for couples, and the median change in the money metric through the direct reference channel is exactly 0 euros.

Changes in earning opportunities nevertheless change the measure, because they change what the household attains: the median total change is -17.43 euros per month for single adults and 110.53 for

couples. The correct statement is therefore narrow and we make only it: *the reference is directly pay-neutral, and earning opportunities reach the measure through the attained evaluation*. A total change that travels through attainment does not establish that the deterministic independence-of-pay axiom fails for this stochastic functional. That would require fixing the primitives the axiom holds fixed and proving the property, which we have not done and do not claim.

4.3 Interpersonal comparison and the unit

The measure is defined at the household. Comparing households of different size requires an equivalence scale, which is a normative choice and not an estimate. We report every result on two bases: a raw household basis, and an equivalized basis using the modified OECD scale. We never pool the two household types into one distribution, because the two applications carry different reference constructions and the levels are not comparable; every share below is a share of the baseline inequality of its own population.

4.4 The structural decomposition

Let $X_i = (P_i, A_i, B_i, D_i)$ collect the four structural inputs for household i :

- P_i the arguments of the preference index: the leisure-weight covariates and the sex- or spouse-specific preference block;
- A_i the arguments of job access: the employment index covariates and the occupation access table;
- B_i the arguments of earning opportunities: the covariates entering the offered-wage location;
- D_i the household budget inputs: non-labour resources, the roster and needs.

Let $\mathcal{I}$ be an inequality index and $W_i(\cdot)$ the money metric of the previous subsection, so that the baseline is $I_\emptyset = \mathcal{I}\{W_i(X_i)\}_{i=1}^N$. For each factor define a **structural equalization operator** T_P, T_A, T_B, T_D , which replaces that factor's arguments across all households by a common reference profile and leaves the estimated coefficients in place. For a coalition $S \subseteq \{P, A, B, D\}$ let $X^S = T_S(X)$ be the state in which exactly the factors in S are equalized, and set

$$I_S = \mathcal{I}\{W_i(T_S X)\}_{i=1}^N, \quad v(S) = I_\emptyset - I_S.$$

v is a cooperative game on four players: the worth of a coalition is the inequality it removes when its factors are equalized together. Two points about T_S are part of the economics and not of the notation. First, T_S is a single simultaneous substitution map, not an ordered product $\prod_{k \in S} T_k$: a product is not well defined unless the operators commute on the permitted objects, and the budget operator does not commute with the others, since it reprices. Second, an operator changes a *pathway*, not every occurrence of a raw characteristic. Education enters both the offered-wage location and the local-market lookup; T_B substitutes the first and leaves the second in place.

Table 5: The four structural equalization operators. Each operator replaces the arguments of one structural pathway with a common reference profile and leaves the estimated coefficients in place; it does not equalize every occurrence of a raw characteristic. Education, for example, enters both the wage location and the local-market lookup, and only the named pathway is substituted. Operators are applied as one simultaneous substitution map, not as an ordered product.

Operator	What is replaced	What is retained	Repricing
T_P preferences	Singles: the arguments of the leisure weight and the complete reference-sex leisure block. Couples: the medoid spouse arguments, with own spouse coefficients retained	Budget roster, resources, access and wage pathways of the same characteristics	No: pure utility shifters do not change the budget

Operator	What is replaced	What is retained	Repricing
T_A job access	The arguments of the employment index and the occupation access table	Preferences, wage location, budget inputs; sex-specific occupation coefficients	No: the priced jobs are unchanged
T_B earning opportunities	The arguments of the offered-wage location: education shares and experience moments, with squares recomputed rather than averaged	The estimated wage coefficients and dispersion; the preference and access pathways of the same characteristics	No on a common priced node set: the change is in the density over nodes
T_D resources and needs	The non-labour budget inputs, the household roster and the needs profile	Every non-budget structural pathway	Yes: the household budget is recomputed through the tax-benefit system

Welfare and inequality are then recomputed from the model for all sixteen coalitions. There is no linearisation and no re-estimation: coefficients are held at their estimates throughout, and only the arguments move. The budget operator additionally reruns the household through the tax-benefit system, so its counterfactual consumption is priced rather than scaled.

The interactions are allocated with the **grouped Shapley–Owen–Shorrocks** rule for the partition $\{\{P\}, \{A, B, D\}\}$: the Owen value (Owen, 1977) for a game with a priori unions, applied to v in the form Shorrocks (2013) sets out for distributional analysis and Audoly et al. (2025) document for grouped nonlinear decompositions. The grouping is chosen because the paper’s question is first about preferences against circumstances and only then about which circumstance; the lower-level subdivision of $\{A, B, D\}$ is allocated within the union. The allocation is exhaustive: the contributions sum to $I_\emptyset - I_{\{P, A, B, D\}}$, and $I_{\{P, A, B, D\}}$ is zero to numerical precision. We verify that closure rather than imposing it; residuals are reported in Section 5.

5 Empirical results

5.1 Behavioural estimates

The estimated model has 41 free coordinates for single adults, of which 40 are interior, and 47 for couples, all interior. The criterion is 6253.463 and 10283.034 respectively. Across ten terminal paths from five starts under two polishing contracts, the criterion varies by 1.1e-10 for single adults and 1.3e-09 for couples, and the smallest eigenvalue of the exact Hessian on the interior block is 0.1669 and 0.0771. One single-adult coordinate, the female age-square term in the leisure weight, sits at a box endpoint; its interval is reported under the active-set convention in the appendix and it is excluded from the interior curvature.

Table 6: The preference block. Cluster-robust standard errors in parentheses, clustered on the household. “Restricted” marks a coordinate held at a value by the specification, not an estimated zero; “at bound” marks an estimate at a box endpoint, whose interval is reported under the active-set convention in the appendix. The consumption weight is common within a population and enters as $\beta_c \log(C/\lambda_c)$.

Coefficient	Single men	Single women	Couple men	Couple women
Leisure weight, intercept $\beta_{\ell 0}$	8.5199 (3.5480)	5.8683 (2.1346)	3.9814 (0.7466)	13.1029 (4.7760)
Leisure weight, age $\beta_{\ell a}$	1.4820 (1.0236)	0.0678 (0.4792)	-0.0080 (0.0287)	-0.1228 (0.1260)
Leisure weight, age squared $\beta_{\ell a^2}$	0.7783 (0.7732)	1.0000 (at bound)	0.0065 (0.0030)	0.0073 (0.0103)
Leisure weight, children $\beta_{\ell n}$	0 (restricted)	0.1666 (0.4422)	0 (restricted)	-0.3055 (1.1472)
Leisure curvature θ_ℓ	-1.6263 (0.3301)	-0.9274 (0.2161)	-0.9761 (0.1357)	-1.6863 (0.2458)

Coefficient	Single men	Single women	Couple men	Couple women
Consumption weight β_c	2.0387 (0.2917)	2.0387 (0.2917)	2.1017 (0.2939)	2.1017 (0.2939)
Consumption curvature θ_c	0 (restricted)	0 (restricted)	0 (restricted)	0 (restricted)
Direct cross-leisure term	Not applicable	Not applicable	0 (restricted)	0 (restricted)

The consumption weight is 2.0387 with a cluster-robust standard error of 0.2917 for single adults and 2.1017 with 0.2939 for couples. Two things follow. Because the estimate exceeds one, the money metric of Section 4 is a power mean of order above one rather than an arithmetic mean. And because utility is on the unit-scale shock, one natural unit of the index corresponds to multiplying consumption by $\exp(1/\beta_c)$, a factor of 1.633 for single adults and 1.609 for couples. That is a proportional statement: there is no single euro value of a unit of utility independent of the consumption at which it is evaluated.

Table 7: The opportunity block: employment access, the hours density and occupation access. Cluster-robust standard errors in parentheses. The employment index multiplies the working indicator; the omitted references are region 1, thinly populated areas, the residual hours set of total width 26.5 hours per week, and occupation group 1. The singles occupation column reports the female coordinates; the male coordinates are in the appendix table. Hours coefficients are common across the two singles sex groups and spouse-specific for couples.

Coefficient	Singles	Couple men	Couple women
Employment constant β_E	-3.1735 (0.4053)	-2.1233 (0.3195)	-2.8148 (0.3021)
Group unemployment rate $\beta_{E,gsur}$	-1.4422 (0.2356)	-1.1924 (0.1557)	shared with the man
Densely populated $\beta_{E,u}$	-0.0288 (0.2209)	-0.1823 (0.1657)	shared with the man
Intermediate density $\beta_{E,m}$	0.0641 (0.2614)	-0.4174 (0.1861)	shared with the man
Region 2 $\beta_{E,2}$	-0.3780 (0.3292)	-0.1546 (0.2442)	shared with the man
Region 3 $\beta_{E,3}$	-0.1117 (0.3891)	0.0873 (0.2808)	shared with the man
Region 4 $\beta_{E,4}$	-0.8218 (0.3810)	0.0150 (0.3068)	shared with the man
Region 5 $\beta_{E,5}$	-0.5162 (0.3274)	-0.1455 (0.2555)	shared with the man
Region 6 $\beta_{E,6}$	-0.7222 (0.3528)	-0.2844 (0.2748)	shared with the man
Region 7 $\beta_{E,7}$	-0.5374 (0.3503)	-0.1396 (0.2651)	shared with the man
Region 8 $\beta_{E,8}$	-0.4512 (0.3401)	-0.1659 (0.2660)	shared with the man
Hours band: Part-time lower, [17.5, 21.5)	0.0298 (0.1972)	-1.1094 (0.3791)	-0.5173 (0.1629)
Hours band: Part-time upper, [28.5, 30.5)	0.7080 (0.1928)	0.0835 (0.2378)	1.1586 (0.1214)
Hours band: Narrow full-time, [33.5, 36.5)	2.0656 (0.0894)	2.2888 (0.0821)	2.0284 (0.0682)
Hours band: Full-time upper, [36.5, 40.5]	1.9387 (0.0997)	2.3801 (0.0866)	1.6825 (0.0798)
Hours band: Long hours, [44.5, 70]	-0.0834 (0.1772)	0.6977 (0.1335)	-0.3109 (0.1615)
Occupation 2 ξ_2	0.1135 (0.1374)	-1.5097 (0.0967)	0.2074 (0.0891)
Occupation 3 ξ_3	-0.4060 (0.1440)	-2.2532 (0.1244)	-0.1804 (0.0923)
Occupation 4 ξ_4	0.5632 (0.1225)	0.1594 (0.0618)	0.8159 (0.0799)

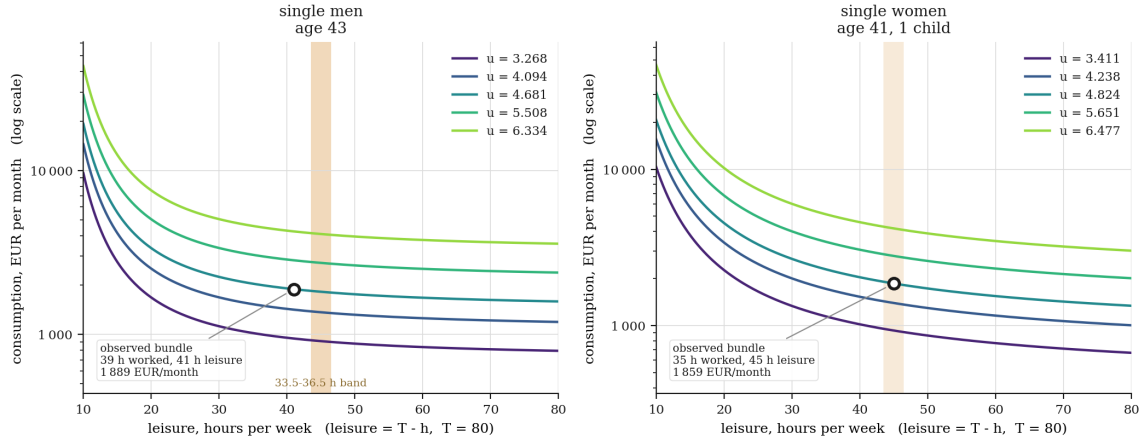


Figure 5: Indifference curves in consumption and leisure, single-adult households, at one representative household of each sex. The budget set, the opportunity density and the taste shock are not drawn, so a curve is not a set of attainable bundles. The consumption axis is logarithmic; under log consumption every finite utility target is attainable at a finite positive consumption, which may lie outside the plotted range.

Table 8: The wage block and the consumption weight. The offered log wage is normal with mean $\mu_i(k)$ and dispersion σ , truncated to $[2, 590]$ EUR/hour and renormalized on that support. Experience enters in units of twenty years. Slopes and σ are common across the two singles sex groups and across spouses within couples; the singles and couples models are estimated separately, so the two columns are not restricted to agree.

Coefficient	Singles	Couples
Intercept β_{w0}	2.0138 (0.0561)	2.0480 (0.0335)
Low education β_{wL}	0.0566 (0.0372)	-0.0433 (0.0217)
High education β_{wH}	0.1491 (0.0308)	0.1817 (0.0197)
Potential experience β_{wx}	0.2408 (0.0862)	0.5643 (0.0546)
Experience squared β_{wx^2}	-0.0282 (0.0390)	-0.1598 (0.0243)
Occupation 2 δ_2	-0.0347 (0.0399)	-0.0740 (0.0231)
Occupation 3 δ_3	0.0622 (0.0383)	0.0394 (0.0228)
Occupation 4 δ_4	0.2785 (0.0368)	0.2146 (0.0223)
Log-wage dispersion σ	0.3815 (0.0133)	0.3631 (0.0072)
Consumption weight β_c	2.0387 (0.2917)	2.1017 (0.2939)

The year indicators carried by the pooled specification are identically zero on this single-year sample and contribute nothing; they are restrictions, not estimates. The singles and couples models are estimated separately, so their wage blocks are not restricted to agree, and the difference between the two experience profiles should not be read as a test.

The indifference curves and the marginal rate of substitution are the economics; the coefficients are coordinates. Note that the consumption axis is logarithmic and the curves are drawn over the plotted range only: under $L(\ell) + \beta_c \log(C/\lambda_c)$, any finite utility target at positive leisure is reached at the finite positive consumption $C = \lambda_c \exp\{(\bar{u} - L(\ell))/\beta_c\}$, which is strictly positive for every finite target. A curve leaving the frame has left the plotting range, not the domain of the model, and nothing here is economically infeasible.

5.2 Fit

The fit reported here is a population prediction, computed by integrating the estimated model over the opportunity distribution and the taste shocks. It is not a sampled-menu choice probability and it is not an in-sample fitted value.

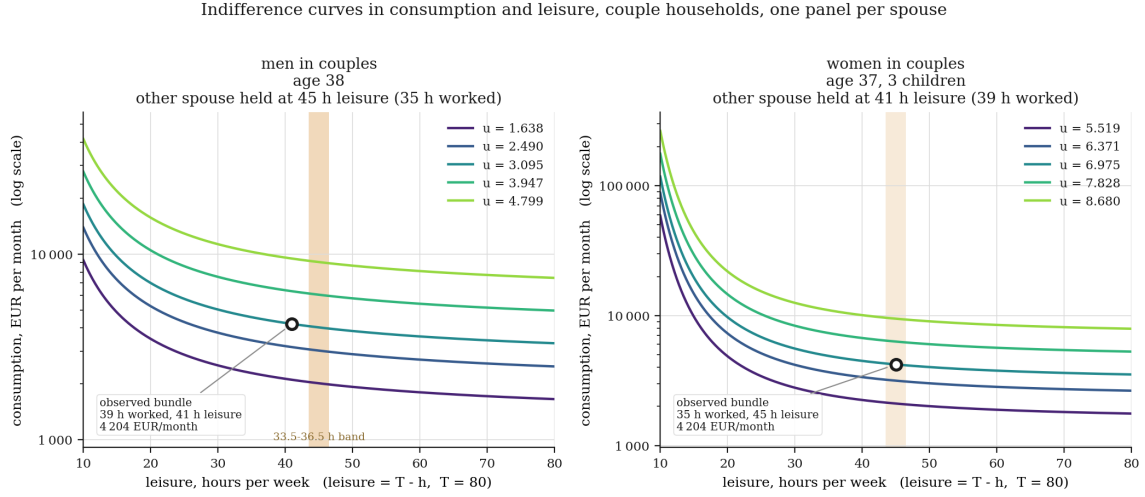


Figure 6: Indifference curves in consumption and each spouse's leisure, couple households, with the partner's leisure held at its observed value. Conditional slices of a joint object, not attainable sets.

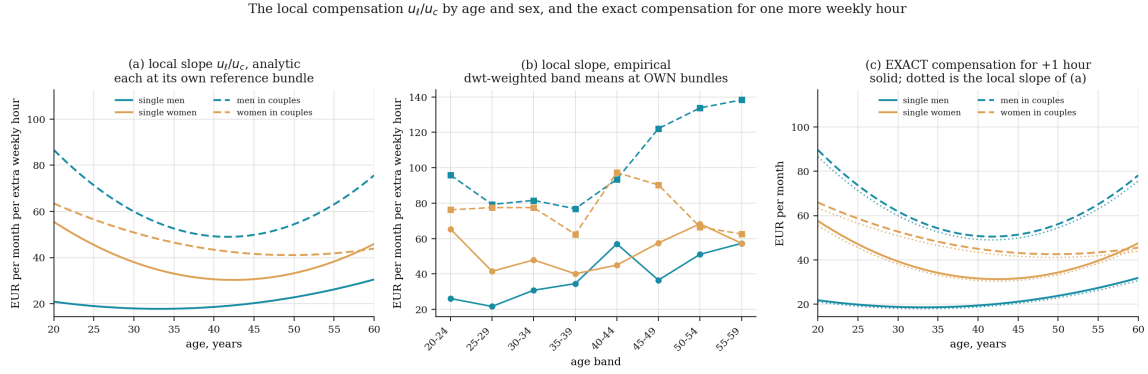


Figure 7: Marginal rate of substitution between leisure and consumption, by age and sex, in euros per month per additional recurring weekly hour of leisure. A compensation along an indifference curve, not a behavioural response to a wage change.

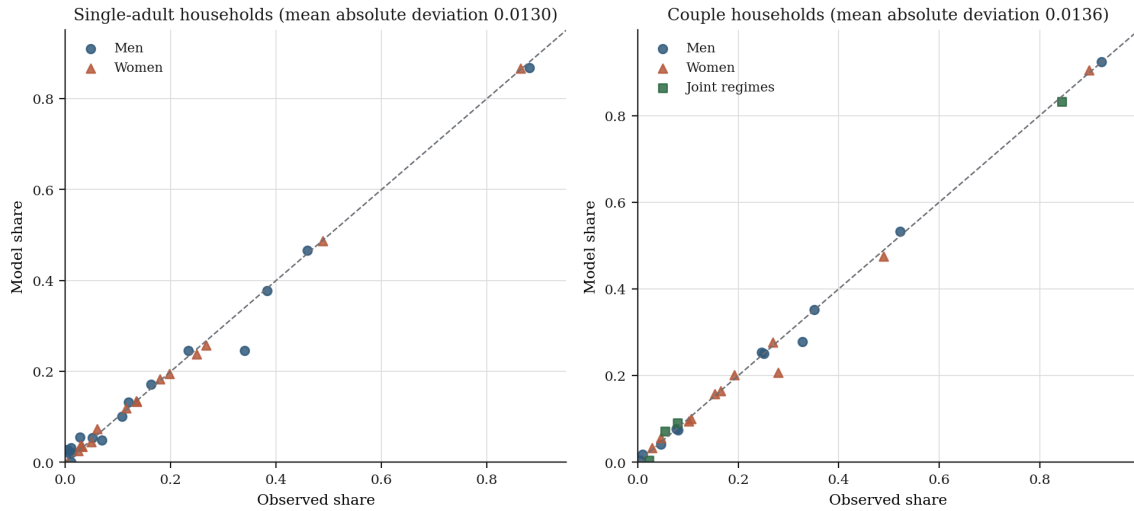


Figure 8: Observed against model population shares, margin by margin, for both estimated specifications. Model shares are population predictions computed by direct integration over the estimated opportunity distribution, not sampled-menu choice probabilities.

Table 9: Observed against model population shares, singles. The model column is a population prediction computed by direct integration over the estimated opportunity distribution and the taste shocks; it is not a sampled-menu choice probability and not an in-sample fitted value. The denominator column states the population each share is taken over.

Unit	Moment	Observed	Model	Absolute deviation	Denominator
Man	Employment	0.8805	0.8675	0.0130	households
Man	Hours: non-employment	0.1195	0.1325	0.0130	households
Man	Hours: [5, 10)	0.0109	0.0000	0.0109	households
Man	Hours: [10, 17.5)	0.0075	0.0198	0.0123	households
Man	Hours: [17.5, 21.5)	0.0118	0.0215	0.0097	households
Man	Hours: [21.5, 28.5)	0.0278	0.0548	0.0269	households
Man	Hours: [28.5, 30.5)	0.0039	0.0285	0.0246	households
Man	Hours: [30.5, 33.5)	0.0107	0.0315	0.0208	households
Man	Hours: [33.5, 36.5)	0.2329	0.2457	0.0128	households
Man	Hours: [36.5, 40.5]	0.3400	0.2452	0.0948	households
Man	Hours: (40.5, 44.5)	0.0691	0.0488	0.0203	households
Man	Hours: [44.5, 70]	0.1626	0.1717	0.0091	households
Man	Occupation group 1	0.3823	0.3774	0.0049	workers
Man	Occupation group 2	0.1069	0.1014	0.0054	workers
Man	Occupation group 3	0.0516	0.0545	0.0029	workers
Man	Occupation group 4	0.4593	0.4667	0.0074	workers
Man	Mean log wage	2.6876	2.6309	0.0567	workers
Woman	Employment	0.8642	0.8665	0.0024	households
Woman	Hours: non-employment	0.1358	0.1335	0.0024	households
Woman	Hours: [5, 10)	0.0042	0.0000	0.0042	households
Woman	Hours: [10, 17.5)	0.0243	0.0254	0.0010	households
Woman	Hours: [17.5, 21.5)	0.0325	0.0346	0.0022	households
Woman	Hours: [21.5, 28.5)	0.0603	0.0737	0.0134	households
Woman	Hours: [28.5, 30.5)	0.0297	0.0355	0.0058	households
Woman	Hours: [30.5, 33.5)	0.0290	0.0367	0.0077	households
Woman	Hours: [33.5, 36.5)	0.2672	0.2581	0.0091	households

Unit	Moment	Observed	Model	Absolute deviation	Denominator
Woman	Hours: [36.5, 40.5]	0.2493	0.2386	0.0107	households
Woman	Hours: (40.5, 44.5)	0.0492	0.0446	0.0045	households
Woman	Hours: [44.5, 70]	0.1159	0.1192	0.0034	households
Woman	Occupation group 1	0.1795	0.1832	0.0036	workers
Woman	Occupation group 2	0.1976	0.1946	0.0031	workers
Woman	Occupation group 3	0.1349	0.1348	0.0002	workers
Woman	Occupation group 4	0.4879	0.4875	0.0004	workers
Woman	Mean log wage	2.5968	2.6207	0.0238	workers

Table 10: Observed against model population shares, couples. The model column is a population prediction computed by direct integration over the estimated opportunity distribution and the taste shocks; it is not a sampled-menu choice probability and not an in-sample fitted value. The denominator column states the population each share is taken over.

Unit	Moment	Observed	Model	Absolute deviation	Denominator
Household	Joint regime neither works	0.0222	0.0043	0.0178	households
Household	Joint regime man only	0.0794	0.0909	0.0115	households
Household	Joint regime woman only	0.0547	0.0723	0.0176	households
Household	Joint regime both work	0.8437	0.8324	0.0113	households
Man	Employment	0.9231	0.9234	0.0002	households
Man	Hours: non-employment	0.0769	0.0766	0.0002	households
Man	Hours: [17.5, 21.5)	0.0059	0.0046	0.0013	households
Man	Hours: [28.5, 30.5)	0.0095	0.0189	0.0094	households
Man	Hours: [33.5, 36.5)	0.2515	0.2507	0.0008	households
Man	Hours: [36.5, 40.5]	0.3274	0.2780	0.0494	households
Man	Hours: [44.5, 70]	0.2461	0.2531	0.0070	households
Man	Occupation group 1	0.3519	0.3516	0.0003	workers
Man	Occupation group 2	0.0797	0.0741	0.0056	workers
Man	Occupation group 3	0.0464	0.0412	0.0052	workers
Man	Occupation group 4	0.5220	0.5331	0.0111	workers

Unit	Moment	Observed	Model	Absolute deviation	Denominator
Man	Mean log wage	2.7687	2.7054	0.0633	workers
Woman	Employment	0.8984	0.9047	0.0063	households
Woman	Hours: non-employment	0.1016	0.0953	0.0063	households
Woman	Hours: [17.5, 21.5)	0.0283	0.0335	0.0052	households
Woman	Hours: [28.5, 30.5)	0.0462	0.0572	0.0110	households
Woman	Hours: [33.5, 36.5)	0.2687	0.2767	0.0080	households
Woman	Hours: [36.5, 40.5]	0.2793	0.2072	0.0721	households
Woman	Hours: [44.5, 70]	0.1061	0.1002	0.0060	households
Woman	Occupation group 1	0.1653	0.1648	0.0005	workers
Woman	Occupation group 2	0.1922	0.2017	0.0095	workers
Woman	Occupation group 3	0.1533	0.1583	0.0050	workers
Woman	Occupation group 4	0.4892	0.4752	0.0140	workers
Woman	Mean log wage	2.6072	2.6319	0.0247	workers

The mean absolute deviation over all population moments is 0.0130 for single adults and 0.0136 for couples, but the informative content is margin by margin. Employment and the joint participation regimes are matched closely; the narrow full-time band around thirty-five hours is matched closely for both sexes; occupation shares given work are matched to within about one percentage point. The largest single discrepancy in both models is the upper full-time band [36.5, 40.5], which the model under-predicts by about nine percentage points for single men and seven for women in couples, with the mass appearing in the adjacent bands. The model also assigns no mass to observed hours below ten, where the observed shares are around one per cent. Both are limitations of the banded hours density and we report them rather than summarise them away.

Table 11: The estimated model against two re-estimated common-opportunity benchmarks, on the same 1,540 single-adult households, the same sampled alternatives and the same criterion. A better maximized criterion does not by itself establish that a mechanism has been identified; the margin-by-margin comparison in the fit table and figure is the substantive evidence.

Specification	Free coordinates	Criterion	Difference	Population fit
Latent jobs with household-specific opportunities	41	6253.463	–	0.0130
Benchmark A: common opportunity distribution, preferences re-estimated	10	6403.974	+150.51	0.0270

Specification	Free coordinates	Criterion	Difference	Population fit
Benchmark B: common opportunity shape, employment and hours moved into utility and re-estimated	16	6395.108	+141.64	0.0261

The two re-estimated common-opportunity benchmarks are worse by 141.64 and by a larger margin, on the same households, the same sampled alternatives and the same criterion. The comparison is a nested one in the sense that the benchmarks restrict the opportunity block and re-estimate everything else, but we do not convert it into a formal test, because the sampled-alternative criterion is not the likelihood of the observed data and the conditions for a likelihood-ratio distribution are not established here. What the table does support is that the deterioration is concentrated where the opportunity block does its work: the population fit worsens from 0.0130 to 0.0270 and 0.0261, and the occupation margins deteriorate by an order of magnitude, from about half a percentage point to eleven. A better criterion does not by itself establish that a mechanism has been identified.

5.3 Money-metric well-being

Table 12: The distribution of money-metric well-being, and the priced disposable income it replaces. Levels are in euros per month of equivalent flat consumption, weighted. Levels are not comparable between the two household types: each type carries its own reference construction, and the Gini of the money metric and the Gini of income are not two estimates of one quantity.

Population	Basis	Mean	p10	Median	p90	Gini
Single-adult	Household	4,354	2,643	4,139	6,219	0.1936
Single-adult	Equivalized	3,957	2,286	3,722	5,819	0.2089
Single-adult	Priced disposable income	–	–	–	–	0.2533
Couple	Household	1,839	1,353	1,767	2,326	0.1325
Couple	Equivalized	972	721	928	1,229	0.1274
Couple	Priced disposable income	–	–	–	–	0.2310

The money metric is less unequal than the priced disposable income it replaces, for both household types: the Gini falls from 0.253 to 0.194 for single adults and from 0.231 to 0.132 for couples. The two Ginis are not two estimates of one quantity. The income Gini describes an outcome; the welfare Gini describes an ex-ante monetary level built from each household’s own preferences and own reachable jobs, and the difference between them is not a correction but a change of object.

Levels differ from income in opposite directions for the two types, and the levels are not comparable between them: each application carries its own reference construction, with a female-primary reference block for single adults and a medoid-spouse reference for couples, and the couples index adds two leisure terms. We use levels only within type and offer no mechanism for the between-type difference in levels.

5.4 The decomposition

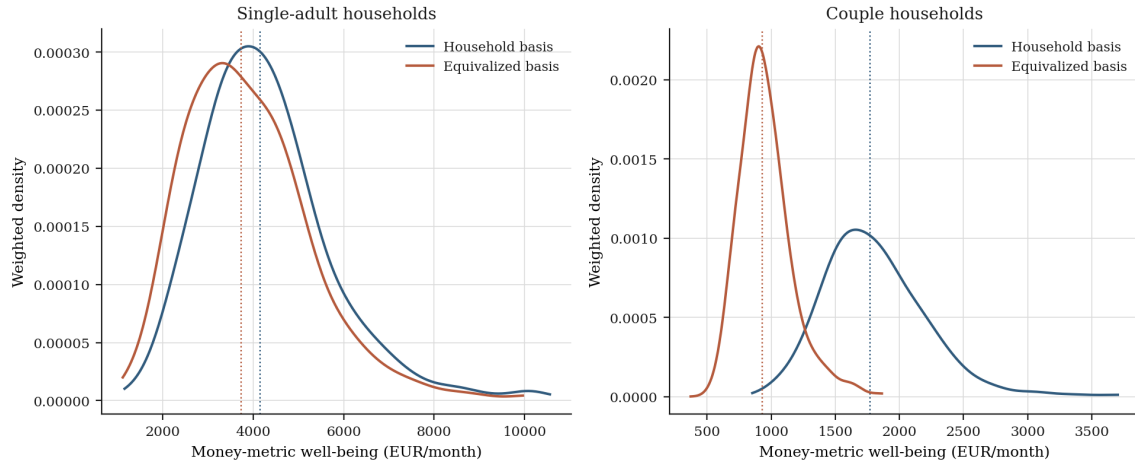


Figure 9: The distribution of money-metric well-being, by household type and basis, with weighted medians marked. Levels are not comparable across the two panels: each household type carries its own reference construction.

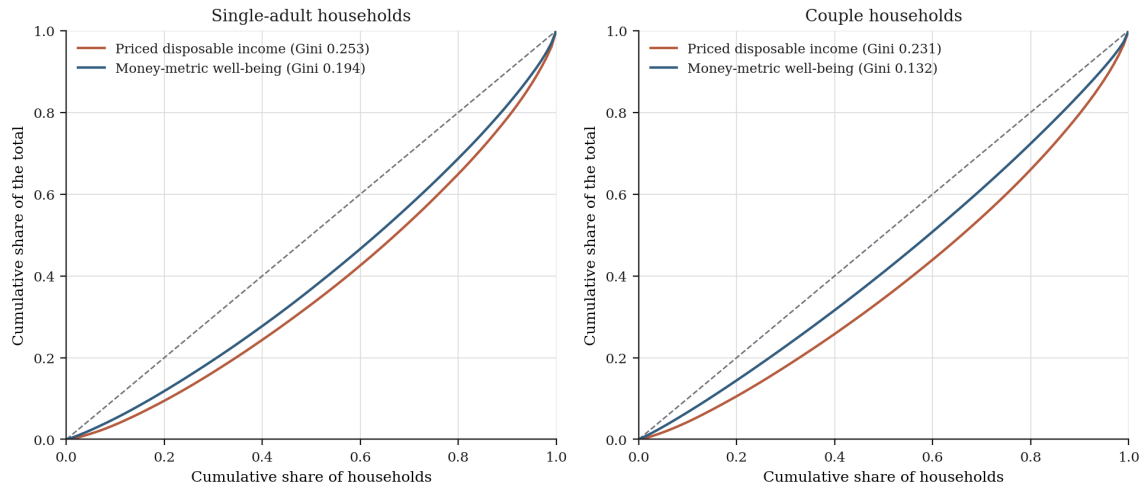


Figure 10: Lorenz curves of money-metric well-being and of the priced disposable income it replaces, over the same households and the same weights. The two curves answer different questions about the same households; their Gini values are not two estimates of one quantity.

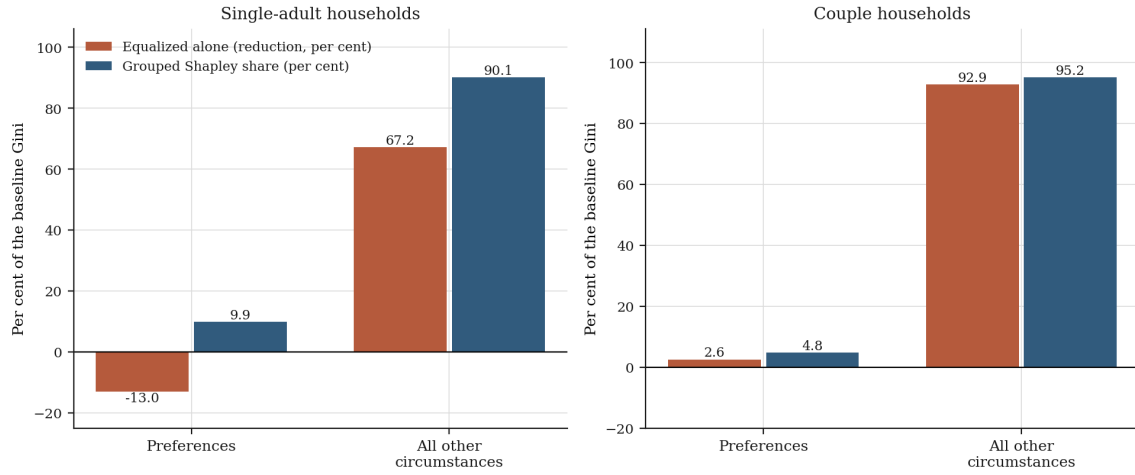


Figure 11: Equalizing one group alone against the grouped Shapley share. For single-adult households, equalizing preferences alone raises the Gini while the preference share is positive; the share averages marginal contributions over coalition orders and the one-factor effect does not.

Table 13: The four coalition states of the two-group game and the corresponding one-factor effects. Weighted Gini of the money metric on the raw household basis, with the female-primary reference for single-adult households. The fully common state is zero to the precision reported in the text: that is a tested property of the game, not an imposed constraint. A positive percentage in the last two rows is a rise in inequality.

State	Single-adult	Couple
Own preferences, own circumstances	0.193596	0.132465
Common preferences, own circumstances	0.218795	0.129046
Own preferences, common circumstances	0.063459	0.009427
Common preferences, common circumstances	0.000000	0.000000
<i>Preferences equalized alone: change in the Gini (per cent)</i>	+13.02	-2.58
<i>All other circumstances equalized alone: change (per cent)</i>	-67.22	-92.88

The four states are the coalition values of the two-group game. Reading them directly gives the one-factor effects. For single adults, equalizing preferences alone *raises* the Gini by 13.0 per cent, from 0.193596 to 0.218795; equalizing all non-preference circumstances alone lowers it by 67.2 per cent. For couples, equalizing preferences alone lowers the Gini by 2.6 per cent and equalizing all other circumstances lowers it by 92.9. The fully common state is zero to $1.3e-15$ in index units, which is a tested property of the game and not an imposed constraint.

The allocated shares are not those numbers. An allocated share is an average of marginal contributions over the orders in which coalitions can form; the one-factor effect is the contribution in exactly one order. For single adults the two even differ in sign, and the arithmetic of the game explains why. With four states and $I_{11} = 0$, the two-group closure gives

$$2C_P = I_{00} - I_{10} + I_{01}.$$

Since $I_{01} \geq 0$ for a nonnegative index, a negative C_P forces $I_{00} - I_{10} < 0$, so a negative allocated share does imply that preference-only equalization raises inequality. The converse does not hold, and the single-adult Gini is the counterexample: $C_P > 0$ while $I_{10} > I_{00}$. This is a consequence of the closure of *this* exhaustive two-group game with a nonnegative index; it is not a general property of a Shapley allocation.

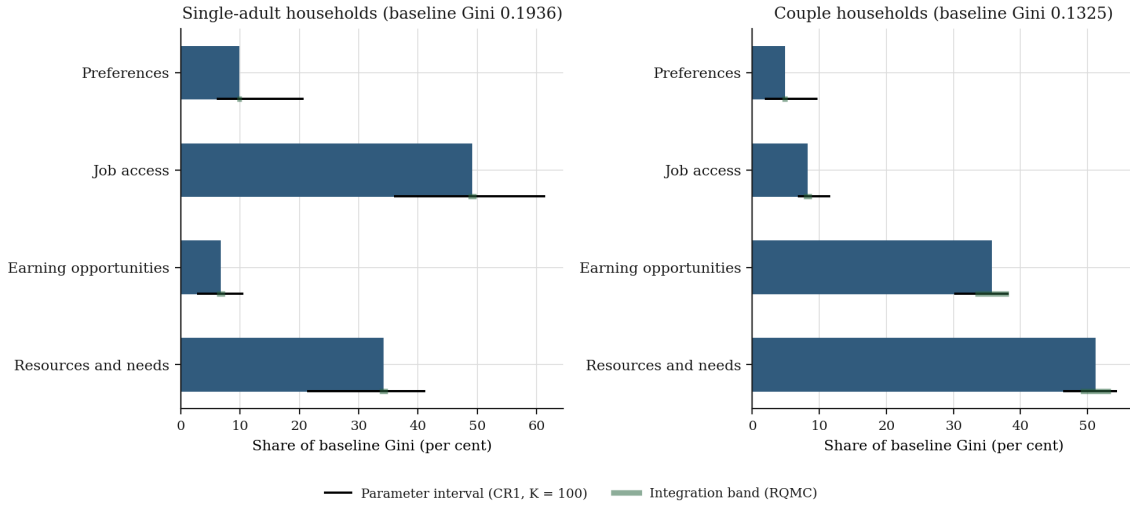


Figure 12: The grouped attribution, signed, in per cent of each population’s own baseline Gini. The black bar is the 95 per cent cluster-robust parameter interval; the shaded bar is the RQMC integration band. They measure different things and are never combined.

Table 14: Grouped attribution of well-being inequality. Contributions in Gini points beside the share of the baseline Gini of the same population, per cent, with the 95 per cent cluster-robust parameter interval in brackets from 100 draws. Shares are taken against the baseline of the same population and are not comparable as levels across the two populations. The parameter interval and the RQMC integration band measure different things and are never combined; the integration band is reported separately in the text and in the figure. The couples subdivision of resources against household composition is not available and is reported jointly.

Component	Singles: Gini points	Singles: share [95 per cent]	Couples: Gini points	Couples: share [95 per cent]
Preferences	0.019130	9.88 [6.1, 20.7]	0.006423	4.85 [1.8, 9.7]
Job access	0.095181	49.16 [36.0, 61.5]	0.010901	8.23 [6.8, 11.6]
Earning opportunities	0.013079	6.76 [2.7, 10.5]	0.047313	35.72 [30.2, 38.3]
Market opportunities (A + B)	0.108260	55.92 [43.0, 66.4]	0.058215	43.95 [39.7, 47.7]
Resources and needs	0.066206	34.20 [21.3, 41.2]	0.067827	51.20 [46.4, 54.4]
All non-preference circumstances	0.174466	90.12	0.126042	95.15

The headline is the single-adult row. Labour-market opportunities, job access together with earning opportunities, carry 55.92 per cent of the inequality in the money metric; job access alone carries 49.16 per cent, with a 95 per cent cluster-robust parameter interval of 36.0 to 61.5 per cent. Preferences carry 9.88 per cent, earning opportunities 6.76 and household resources and needs 34.20. For couples the ordering is different: earning opportunities carry 35.72 per cent and resources and needs 51.20, while job access carries 8.23. Single adults are access-dominated on the labour-market side; couples are earnings- and resource-dominated, and access matters much less for them.

We report that contrast as a finding and attach no mechanism to it. A second earner is not automatically a buffer: two earnings streams do not mechanically produce less dispersion than one, since that depends on the dependence between them and on how resources pool. Establishing a mechanism would require a separately defined counterfactual, which we have not run. The contrast also cannot be assigned entirely to household type, because the two applications differ in their reference constructions as well; Section 6 reports the reference sensitivity.

The parameter interval and the integration band measure different things and are never combined.

The integration band on the single-adult access share is 48.5 to 49.9 per cent, an order of magnitude narrower than the parameter interval, which is the expected ordering: the numerical integration is the accurate part and the parameters are the uncertain part.

The allocation is exhaustive. The maximum absolute top-level identity residual is 2.8e-17 in index units and the nested residual 5.6e-17. That is a computational validation of the accounting for the declared game. It does not validate the identification of the model or the normative content of the operators.

5.5 Index sensitivity

Table 15: Baseline level and attribution shares under six inequality indices. Every row was recomputed for its own index against its own baseline; no coalition value or share is transferred between indices. Shares within a row sum to one hundred by exhaustiveness. A negative preference share means that equalizing preferences alone would raise that index, which is a property of this exhaustive two-group game with a nonnegative index and is explained in the text.

Population and index	Baseline level	Preferences (per cent)	Access	Earning opportunities	Resources and needs	Access + earnings
Single-adult, Gini	0.193596	9.88	49.16	6.76	34.20	55.92
Single-adult, Atkinson(1)	0.060788	-7.21	55.12	5.18	46.90	60.31
Single-adult, Atkinson(2)	0.115514	-6.83	58.49	4.77	43.57	63.27
Single-adult, GE(0)	0.062714	-7.86	55.42	5.26	47.19	60.67
Single-adult, GE(1)	0.065890	-7.59	49.64	5.22	52.73	54.87
Single-adult, GE(2) $= CV^2/2$	0.153963	-8.52	40.89	4.78	62.86	45.67
Couple, Gini	0.132465	4.85	8.23	35.72	51.20	43.95
Couple, Atkinson(1)	0.030197	2.52	6.38	30.50	60.60	36.88
Couple, Atkinson(2)	0.055239	2.74	6.94	33.10	57.21	40.05
Couple, GE(0)	0.030663	2.55	6.40	30.48	60.57	36.88
Couple, GE(1)	0.034637	2.27	5.70	27.09	64.94	32.79
Couple, GE(2) $= CV^2/2$	0.088142	2.03	4.95	22.87	70.16	27.82

Each index keeps its own coalition values and its own allocation; nothing is transferred between rows. Read carefully, the table supports the following and not more.

- Job access exceeds earning opportunities for single adults under all 6 indices, and earning opportunities exceed access for couples under all 6. This is the robust ordering and it is the one we emphasise.
- Job access alone exceeds resources and needs for single adults under 4 of the six; GE(1) and GE(2) are the exceptions.
- Access and earning opportunities together exceed resources and needs for single adults under 5 of the six, with GE(2) the exception.
- Resources and needs are the largest of the four components for couples under all 6 indices.
- The single-adult preference share is positive for the Gini and negative under the other 5 indices. The preference contribution is therefore *not* a metric-invariant conclusion, and the statement that

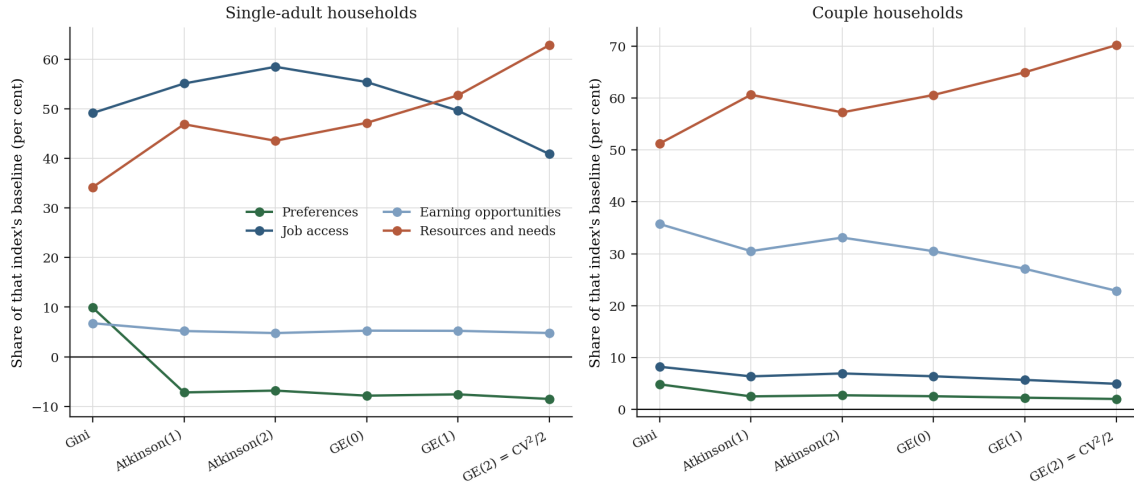


Figure 13: Attribution shares under six inequality indices. Each index keeps its own coalition values and its own allocation. Access exceeds earning opportunities for single adults under all six and the ordering reverses for couples under all six; the singles preference share changes sign outside the Gini.

preferences account for about ten per cent of inequality is a statement about the Gini on the raw basis under the female-primary reference, not a general one.

These are descriptive rankings of point estimates. Parameter intervals are reported for the Gini shares only, and we do not convert any of these orderings into a claim of statistical significance.

6 Sensitivity and limitations

6.1 The normative reference

The reference is a choice, and the results move with it. The single-adult decomposition uses a female-primary reference block; under the alternative male structural-zero reference the shares become 10.17 per cent for preferences, 51.55 for access, 7.56 for earning opportunities and 30.72 for resources and needs. The ordering is unchanged and the magnitudes move by a few percentage points. That sensitivity is real and belongs with the result; it is a property of the normative construction, not a numerical instability.

Equivalization is a second normative choice of the same kind. On the equivalized basis the single-adult access share falls and the resources-and-needs share rises, and the couples pattern moves in the same direction. Both bases are reported throughout; neither is the correct one, and the choice between them is not settled by the data.

6.2 Two open econometric questions

The sample screens on the observed hours and wage of employed deciders. The estimation sample is therefore selected on an outcome of the process being modelled, and the conditional sampled-set likelihood, which is the right object given a sampled choice set, is not automatically the right object given an outcome-selected sample. We have not established which correction the screen requires, or that none is required. This is an econometric question and not a presentational one, and it is not resolved by the diagnostics reported above.

Separately, the sampled-set probability is derived by conditioning on a labelled collection of slots. Small observed cross-coordinate correlations and a multiplicity distribution consistent with independent draws are diagnostics; they do not by themselves establish the sampling law that the derivation assumes. Both questions are recorded here as open.

6.3 The bridge to the compensation-side reference

The flat-consumption reference of Section 4 is one member of a family. A natural comparison is the reference that flattens consumption only at the non-work bundle, which sits on the compensation side

of the same family. If non-work maximizes the non-consumption index and the opportunity density integrates to one on the domain the reference mass uses, then $H_i \leq e^{L_i(o)}$, so the gap $\Delta_i = L_i(o) - \log H_i$ must be non-negative for every household and the ratio of the two measures is bounded on one side.

The first premise holds for every household in both samples. The second does not: the opportunity object entering the reference mass is an unnormalized index whose median mass on that domain is 0.077283 for single adults and 147.963 for couples. Omitting that mass makes the comparison depend on an arbitrary multiplicative scale, and it is the reason earlier signed gaps had the wrong sign for couples and happened not to reverse for single adults. Both earlier levels are withdrawn.

Table 16: The bridge between the two monetary references, after the premise audit. The second column reports whether non-work maximizes the non-consumption index for every household; the third reports the median mass of the opportunity kernel on the domain the reference integral uses. That mass is not one, which is the premise that failed and the reason the earlier signed gaps were scale artefacts. The remaining columns are the bridge after normalizing the kernel on exactly that domain.

Population	Non-work maximizes L	Median $\int \hat{g}$	Median W^4/W^1	Range	Median Δ (nats)	$\Delta \geq 0$ everywhere
Single-adult	yes	0.077283	0.9843	[0.9169, 0.9990]	0.0322	yes
Couple	yes	147.962795	0.6242	[0.5211, 0.7171]	0.9906	yes

After normalizing the kernel on exactly the domain the reference mass uses, the bridge is well behaved: the gap is non-negative for every household in both samples and the ratio is at most one everywhere, as the premises require. The median ratio is 0.9843 for single adults and 0.6242 for couples. We report the reconciled bridge as a property comparison between two references and do not use it as a competing headline distribution.

6.4 The relative-index companion measure

A third reference, which flattens the sum of a resource base and gross pay rather than disposable consumption, is defined only where the inversion brackets. It brackets for all 1,540 single-adult households with no negative values, and is therefore usable as a single-adult diagnostic. For couples it brackets for 9 of 2,223 households in the theory-strict form, and the household-resource form produces 2,103 negative values, on which relative inequality indices are not defined. We therefore treat it as a single-adult diagnostic and do not force it into the couples application. Failure to bracket is not by itself proof that a measure does not exist; here the three cases we can distinguish are an insufficient upper bracket, an empty positive-consumption domain and a target below the attainable reference range, and we report which applies rather than a single count.

6.5 The consumption normalizer

Table 17: The consumption normalizer. Three constants were in circulation. Under exact log consumption the normalizer enters utility as the alternative-invariant term $-\beta_c \log \lambda_c$, so it cancels from every choice probability and exactly from the money metric. The paper reports the estimation-frame constant throughout.

Panel	Single-adult (EUR/month)	Couple (EUR/month)	Role
Estimation frame, 101 sampled alternatives per household	1,938.238719	4,247.875047	Constant of record
Welfare panel, 2,048 common integration nodes	1,774.518218	3,821.448012	Recomputed over its own rows; inert

Panel	Single-adult (EUR/month)	Couple (EUR/month)	Role
Predecessor frames, before the sample correction	1,911.108058	3,821.448012	Superseded

Three normalizing constants were in circulation between the estimation frames and the welfare panel, because each panel computes the constant as a mean over its own rows. Under exact log consumption the constant is alternative-invariant, so it cancels from every choice probability and exactly from the money metric; the deviation over four widely separated values is $2.7\text{e-}15$. No reported quantity changes. The paper reports the estimation-frame constant throughout, and the discrepancy is recorded here rather than silently harmonised.

6.6 What is not established

The decomposition is an accounting of a structural model under declared operators. It is not a causal analysis. No regional, educational or occupational effect reported here is identified as a causal effect, and the exhaustiveness of the allocation validates the accounting for the declared game, not the identification of the model or the normative content of the operators.

The couples subdivision of resources against household composition is not available: it requires two repriced counterfactual panels that have not been run, and we report the two jointly at 51.20 per cent rather than imputing a split. The parameter intervals cover the Gini shares; the other five indices are reported as point estimates.

7 Conclusion

Observed hours and earnings do not say whether a household chose its position or settled for it, and that ambiguity is not a nuisance for welfare measurement: it is the substance of it. This paper takes the ambiguity seriously in both halves of the problem. On the behavioural side it estimates a model in which the jobs a household can reach and the way it ranks them are identified jointly, with every alternative priced through the tax-benefit system. On the normative side it evaluates well-being with a reference that keeps each household's own preferences and own reachable jobs and removes variation in pay from the reference bundles.

The decomposition then asks how the resulting inequality divides. For French single-adult households, labour-market opportunities carry 55.92 per cent of it and job access alone 49.16 per cent, more than household resources and needs and far more than preferences. For couples the same accounting gives a different answer: earning opportunities and household circumstances dominate and access matters little. The ordering of access against earnings within each type is the part that survives all six inequality indices we report; the preference contribution is not, and changes sign outside the Gini.

Three limits should travel with those numbers. The measure is one member of a family of references and the shares move with the reference, in ways we report rather than resolve. The allocation is exhaustive for the declared game, which validates the accounting and not the identification or the ethics of the operators. And two questions about the observation rule remain genuinely open. What the exercise offers is not a causal account of why opportunities differ, but a disciplined statement of how much of measured well-being inequality is associated with them once preferences, resources and needs are modelled explicitly and the reference is stated.

A Appendix A. The estimated parameter vectors

The tables below report every coordinate of the two estimated parameter vectors, including the coordinates a specification holds fixed. A restricted coordinate contributes no degree of freedom and is not an estimated zero. For the single-adult vector, eight leisure coordinates belonging to the couples block and two year indicators are carried by the pooled specification and are inert on this sample: the year indicators are identically zero on a single-year frame, and the couples leisure coordinates enter no single-adult likelihood term.

One single-adult coordinate, the female age-square term, is at its box endpoint. Under the active-set convention its interval is not reported, the interior curvature is computed after removing it, and the parameter draws used for the welfare intervals hold it at its estimate.

Table 18: The complete singles parameter vector. Standard errors are cluster-robust on the household. A coordinate marked restricted is held by the specification and contributes no degree of freedom.

Coordinate	Estimate	CR1 s.e.	z	Box	Status
beta_10_sm	8.51993	3.54798	2.401	[0.05, 50]	estimated
beta_1_age_sm	1.48195	1.02361	1.448	[-5, 5]	estimated
beta_1_age2_sm	0.778284	0.773158	1.007	[-1, 1]	estimated
theta_1_sm	-1.62627	0.330058	-4.927	[-8, 0.95]	estimated
beta_10_sf	5.86829	2.13459	2.749	[0.05, 50]	estimated
beta_1_age_sf	0.0677662	0.479217	0.141	[-5, 5]	estimated
beta_1_age2_sf	1	—	—	[-1, 1]	at bound
beta_1_nkids_sf	0.166636	0.442236	0.377	[-5, 5]	estimated
theta_1_sf	-0.927353	0.216128	-4.291	[-8, 0.95]	estimated
theta_c_singles	0	—	—	[-8, 0.95]	restricted
beta_10_m	1e-06	—	—	[1e-06, 50]	restricted
beta_1_age_m	-0.0672369	—	—	[-5, 5]	restricted
beta_1_age2_m	0.0877505	—	—	[-1, 1]	restricted
beta_10_f	10.0522	—	—	[0.05, 50]	restricted
beta_1_age_f	-1.78025	—	—	[-5, 5]	restricted
beta_1_age2_f	1	—	—	[-1, 1]	restricted
beta_1_nkids_f	0.58572	—	—	[-5, 5]	restricted
theta_1_f	-2.13174	—	—	[-8, 0.95]	restricted
beta_E	-3.17353	0.405337	-7.829	[-25, 25]	estimated
beta_h_pt1	0.0297672	0.197247	0.151	[-10, 10]	estimated
beta_h_pt2	0.707971	0.192788	3.672	[-10, 10]	estimated
beta_h_ft	1.93873	0.0996968	19.446	[-10, 10]	estimated
beta_h_lh	-0.0833752	0.177201	-0.471	[-10, 10]	estimated
beta_E_gsur	-1.44224	0.235591	-6.122	[-10, 10]	estimated
beta_E_drgn2	-0.377963	0.32925	-1.148	[-10, 10]	estimated
beta_E_drgn3	-0.111713	0.38912	-0.287	[-10, 10]	estimated
beta_E_drgn4	-0.821813	0.380963	-2.157	[-10, 10]	estimated
beta_E_drgn5	-0.516158	0.327362	-1.577	[-10, 10]	estimated
beta_E_drgn6	-0.722159	0.35278	-2.047	[-10, 10]	estimated
beta_E_drgn7	-0.537369	0.350308	-1.534	[-10, 10]	estimated
beta_E_drgn8	-0.45119	0.340074	-1.327	[-10, 10]	estimated
beta_E_y2015	-0.254606	—	—	[-5, 5]	restricted
beta_E_y2017	-0.0694711	—	—	[-5, 5]	restricted
beta_E_drgur	-0.0287689	0.220916	-0.130	[-10, 10]	estimated
beta_E_drgmd	0.064133	0.261352	0.245	[-10, 10]	estimated
beta_occ_2_m	-1.37371	0.158147	-8.686	[-15, 15]	estimated
beta_occ_3_m	-2.10963	0.202248	-10.431	[-15, 15]	estimated
beta_occ_4_m	-0.325897	0.121341	-2.686	[-15, 15]	estimated
beta_occ_2_f	0.113492	0.137399	0.826	[-15, 15]	estimated
beta_occ_3_f	-0.406009	0.144045	-2.819	[-15, 15]	estimated
beta_occ_4_f	0.563216	0.122467	4.599	[-15, 15]	estimated
beta_w0	2.01382	0.0560922	35.902	[-10, 20]	estimated
beta_w_educL	0.0566261	0.0372485	1.520	[-5, 5]	estimated
beta_w_educH	0.149054	0.0308408	4.833	[-5, 5]	estimated
beta_w_pexp	0.24077	0.0862223	2.792	[-1, 1]	estimated
beta_w_pexp2	-0.0281739	0.0389905	-0.723	[-0.1, 0.1]	estimated
sigma	0.381505	0.0132752	28.738	[0.1, 20]	estimated
delta_occ_2	-0.0347155	0.039899	-0.870	[-4, 4]	estimated
delta_occ_3	0.0622139	0.0382749	1.625	[-4, 4]	estimated
delta_occ_4	0.278494	0.0367705	7.574	[-4, 4]	estimated

Coordinate	Estimate	CR1 s.e.	z	Box	Status
beta_h_f35	2.06556	0.0894479	23.092	[-10, 10]	estimated
beta_c	2.03873	0.291729	6.988	[0.05, 50]	estimated

Table 19: The complete couples parameter vector. Standard errors are cluster-robust on the household. A coordinate marked restricted is held by the specification and contributes no degree of freedom.

Coordinate	Estimate	CR1 s.e.	z	Box	Status
beta_10_m	3.98136	0.746569	5.333	[1e-06, 50]	estimated
beta_1_age_m	-0.00799228	0.0287174	-0.278	[-5, 5]	estimated
beta_1_age2_m	0.00648336	0.00297388	2.180	[-1, 1]	estimated
theta_1_m	-0.976109	0.135663	-7.195	[-8, 0.95]	estimated
beta_10_f	13.1029	4.77604	2.743	[0.05, 50]	estimated
beta_1_age_f	-0.122813	0.126048	-0.974	[-5, 5]	estimated
beta_1_age2_f	0.00730709	0.0102503	0.713	[-1, 1]	estimated
beta_1_nkids_f	-0.305471	1.14719	-0.266	[-5, 5]	estimated
theta_1_f	-1.68631	0.245846	-6.859	[-8, 0.95]	estimated
beta_E_m	-2.12326	0.319457	-6.646	[-25, 25]	estimated
beta_E_f	-2.81476	0.302078	-9.318	[-25, 25]	estimated
beta_h_pt1_m	-1.10936	0.379086	-2.926	[-10, 10]	estimated
beta_h_pt1_f	-0.517276	0.162932	-3.175	[-10, 10]	estimated
beta_h_pt2_m	0.0835202	0.237788	0.351	[-10, 10]	estimated
beta_h_pt2_f	1.15858	0.121438	9.541	[-10, 10]	estimated
beta_h_f35_m	2.2888	0.0821348	27.866	[-10, 10]	estimated
beta_h_f35_f	2.02835	0.0682392	29.724	[-10, 10]	estimated
beta_h_ft_m	2.38012	0.0865569	27.498	[-10, 10]	estimated
beta_h_ft_f	1.68251	0.0798241	21.078	[-10, 10]	estimated
beta_h_lh_m	0.69767	0.133487	5.226	[-10, 10]	estimated
beta_h_lh_f	-0.310873	0.161533	-1.925	[-10, 10]	estimated
beta_E_gsur	-1.19243	0.155667	-7.660	[-10, 10]	estimated
beta_E_drgn2	-0.154617	0.24423	-0.633	[-10, 10]	estimated
beta_E_drgn3	0.0872501	0.280833	0.311	[-10, 10]	estimated
beta_E_drgn4	0.0150136	0.306753	0.049	[-10, 10]	estimated
beta_E_drgn5	-0.145472	0.255453	-0.569	[-10, 10]	estimated
beta_E_drgn6	-0.284391	0.274814	-1.035	[-10, 10]	estimated
beta_E_drgn7	-0.139569	0.265133	-0.526	[-10, 10]	estimated
beta_E_drgn8	-0.165927	0.26603	-0.624	[-10, 10]	estimated
beta_E_drgur	-0.182322	0.165665	-1.101	[-10, 10]	estimated
beta_E_drgmd	-0.417356	0.1861	-2.243	[-10, 10]	estimated
beta_occ_2_m	-1.50975	0.0967347	-15.607	[-15, 15]	estimated
beta_occ_3_m	-2.25319	0.124405	-18.112	[-15, 15]	estimated
beta_occ_4_m	0.159368	0.061796	2.579	[-15, 15]	estimated
beta_occ_2_f	0.207379	0.0891054	2.327	[-15, 15]	estimated
beta_occ_3_f	-0.180381	0.0923137	-1.954	[-15, 15]	estimated
beta_occ_4_f	0.81589	0.079925	10.208	[-15, 15]	estimated
beta_w0	2.048	0.0335421	61.058	[-10, 20]	estimated
beta_w_educL	-0.0433046	0.0216643	-1.999	[-5, 5]	estimated
beta_w_educH	0.181652	0.0197059	9.218	[-5, 5]	estimated
beta_w_pexp	0.56435	0.0545928	10.337	[-3, 3]	estimated
beta_w_pexp2	-0.15983	0.0242888	-6.580	[-0.3, 0.3]	estimated
sigma	0.363075	0.0072235	50.263	[0.1, 20]	estimated
delta_occ_2	-0.0740322	0.0231318	-3.200	[-4, 4]	estimated
delta_occ_3	0.0393566	0.0228	1.726	[-4, 4]	estimated
delta_occ_4	0.214559	0.0223344	9.607	[-4, 4]	estimated
beta_c	2.10172	0.293879	7.152	[0.05, 50]	estimated

B Appendix B. Inequality indices and the allocation rule

Table 20: The six inequality indices, defined on a weighted distribution of strictly positive levels with mean μ . Multiplying an index by a positive constant scales its level and its contributions but not its shares, so GE(2) and the squared coefficient of variation are the same game. Atkinson(1) and GE(0) rank any positive distribution identically because $A(1) = 1 - e^{-GE(0)}$; their shares can nevertheless differ, because a nonlinear transformation does not commute with averaging marginal contributions over coalition orders.

Index	Definition
Gini	$\frac{1}{2\mu} \mathbb{E} W - \tilde{W} $, for $W, \tilde{W}$ independent draws from the distribution
Atkinson(1)	$1 - \exp(\mathbb{E} \log W)/\mu$
Atkinson(2)	$1 - (\mathbb{E}[W^{-1}])^{-1}/\mu$
GE(0)	$\mathbb{E} \log(\mu/W)$
GE(1)	$\mathbb{E}[(W/\mu) \log(W/\mu)]$
GE(2) = $CV^2/2$	$\frac{1}{2} \mathbb{E}[(W/\mu)^2 - 1]$

Two relations are worth stating because they explain apparent puzzles in the six-index table. First, multiplying an index by a positive constant scales its level and every contribution but leaves the shares unchanged, so GE(2) and the squared coefficient of variation define the same allocation game and are not two robustness checks. Second, $A(1) = 1 - e^{-GE(0)}$, so Atkinson(1) and GE(0) rank any distribution of positive levels identically. Their allocated shares can nevertheless differ, because a nonlinear transformation of the index does not commute with averaging marginal contributions over coalition orders. That is a property of the allocation rule, not an inconsistency.

The Owen value for the partition $\{\{P\}, \{A, B, D\}\}$ allocates first between the two unions and then within the second, averaging over the orders of unions and, within a union, over the orders of its members. Contributions are signed and are never renormalised to sum to one hundred by construction: they sum to $I_\emptyset - I_{\{P, A, B, D\}}$ because the game closes, and the closure is verified.

C Appendix C. The bridge between the two monetary references

Write $M_i = \int \hat{g}_i d\nu$ for the mass of the opportunity object on the domain the reference integral uses, and o for the non-work bundle. The two references satisfy

$$\beta_c \log(W_i^1/\lambda_c) = \log J_i - \log H_i, \quad \beta_c \log(W_i^4/\lambda_c) = \log J_i - \log M_i - L_i(o),$$

so that $\log(W_i^4/W_i^1) = (\log H_i - \log M_i - L_i(o))/\beta_c$ and $\Delta_i = L_i(o) + \log M_i - \log H_i$. Omitting $\log M_i$ is valid only when $M_i = 1$, and it is not: the audit reports median mass 0.077283 for single adults and 147.963 for couples. The premise that failed is therefore the unit mass of the kernel on that domain, not the maximality of the non-work bundle, which holds for every household in both samples, and not the orientation of the comparison.

After renormalizing the kernel on exactly that domain, the median gap is 0.0322 nats for single adults and 0.9906 for couples, the gap is non-negative for every household, and the ratio lies in $[0.9169, 0.9990]$ for single adults and $[0.5211, 0.7171]$ for couples. The identity above is satisfied to machine precision at the corrected normalization.

A separate point concerns the low-temperature limit. Attained and reference values must be indexed by the same shock scale for the limit to be meaningful. The statement that survives is that the two consistently indexed measures converge as the scale goes to zero; convergence to the value of staying at home evaluated at a different fixed scale is not the same statement and is not made.

D Appendix D. Data, software and replication

Data. French EU-SILC, accessed through Eurostat’s harmonised release, with 2016 survey collection and a 2015 income reference year. The harmonised survey is transformed into an input file for EUROMOD

(Sutherland and Figari, 2013), which applies the French 2015 policy system. Regional labour-market conditions come from the Eurostat regional labour-force series. Access to EU-SILC microdata is granted by Eurostat under its research-access conditions and the data cannot be redistributed with this document.

Sample. 1,540 single-adult and 2,223 couple households, constructed by the screens of Section 2.

Estimator and inference. Conditional likelihood over 100 sampled alternatives per household plus the observed choice, with an out-of-fold proposal correction; cluster-robust standard errors on the household. Optimization is checked by five starts under two polishing contracts and curvature by exact Hessian eigenvalues.

Welfare computation. A common integration panel of 2,048 nodes per household, evaluated under all sixteen coalition states. Integration error is measured by 8 randomized quasi-Monte Carlo scrambles; parameter uncertainty by 100 draws from the cluster-robust covariance rebuilt from the Hessian and the household scores, holding restricted and bound-active coordinates at their estimates. The two are reported separately and are never combined into one band.

Software. Python with JAX (0.10.1) for automatic differentiation, and the EUROMOD connector (0.2.17).

Replication. Code and derived, non-confidential intermediate artefacts will be made available in a public repository on publication. Confidential microdata remain in their permitted environment, so the replication package reproduces every step conditional on authorised access to EU-SILC.

E Scientific history of this result

This section records how the reported specifications and results came to be what they are. It is here so that the argument above does not have to carry it, and so that a reader comparing this version with an earlier one can see what changed and why. Nothing in it is a competing model.

The consumption specification. Earlier drafts reported a specification in which the consumption coefficient was fixed at one as a numeraire, and one in which the single-adult consumption curvature was estimated at $\theta_c \approx 0.168$. Neither is the reported model. The current specification fixes the shock scale, fixes the consumption curvature at exactly zero, and estimates the consumption weight instead. That change improves the criterion by 23.753 log-points for single adults and 15.659 for couples, and it changes the welfare aggregator from an arithmetic mean of reachable consumption to a power mean of order β_c . Every welfare quantity computed under the earlier convention is superseded, and the corresponding figures were regenerated rather than relabelled.

The sample. The predecessor frames contained 1,555 single-adult and 2,275 couple households. Twelve single-adult and fifty-two couple households were removed when the hours support and the occupation mapping were applied to the observed rows, and three further single-adult households were removed because their own observed job prices to non-positive disposable consumption. The current descriptive tables are computed on the resulting 1,540 and 2,223 households, not on the predecessor frames, and the funnel in Section 2 ends where the estimation begins.

The disposable-income convention. Single-adult disposable income was previously aggregated over the decider only; it is now aggregated over all resident household members, which is the convention the couples application always used. The two applications are now on the same accounting convention.

The decomposition result. An earlier draft reported household endowments and needs as the largest component for single adults, and a one-factor figure of about seventy-seven per cent. Both are withdrawn. The corrected computation gives 49.16 per cent to job access against 34.20 to resources and needs, and the corresponding one-factor effect for all non-preference circumstances is 67.2 per cent.

The bridge between references. The signed gap between the flat-consumption reference and the non-work-bundle reference was previously reported with the wrong sign for couples. The premise audit in Section 6 locates the cause in the normalization of the opportunity kernel on the domain the reference mass uses, and the corrected bridge satisfies the inequality the premises require in both samples.

The consumption normalizer. Three values of λ_c were in circulation because different panels recomputed it over their own rows. Under exact log consumption the constant cancels from the money metric, so no result changed; the paper now reports one constant per population.

References

Rolf Aaberge and Ugo Colombino. Structural labour supply models and microsimulation. *International Journal of Microsimulation*, 11(1):162–197, 2018.

- Rolf Aaberge, John K. Dagsvik, and Steinar Strøm. Labor supply responses and welfare effects of tax reforms. *Scandinavian Journal of Economics*, 97(4):635–659, 1995.
- Rolf Aaberge, Ugo Colombino, and Steinar Strøm. Labour supply in Italy: An empirical analysis of joint household decisions, with taxes and quantity constraints. *Journal of Applied Econometrics*, 14(4):403–422, 1999.
- Rolf Aaberge, Ugo Colombino, and Steinar Strøm. Do more equal slices shrink the cake? an empirical investigation of tax-transfer reform proposals in Italy. *Journal of Population Economics*, 17:767–785, 2004. doi: 10.1007/s00148-004-0193-2.
- Richard Audoly, Rory McGee, Sergio Ocampo, and Gonzalo Paz-Pardo. A practitioner’s note on the shapley–owen–shorrocks decomposition. Staff Report 1163, Federal Reserve Bank of New York, 2025.
- Olivier Bargain. Back to the future: decomposition analysis of distributive policies using behavioural simulations. *International Tax and Public Finance*, 19:708–731, 2012. doi: 10.1007/s10797-011-9203-y.
- Olivier Bargain, André Decoster, Mathias Dolls, Dirk Neumann, Andreas Peichl, and Sebastian Siegloch. Welfare, labor supply and heterogeneous preferences: Evidence for Europe and the US. *Social Choice and Welfare*, 41:789–817, 2013.
- Bart Capéau, André Decoster, and Gijs Dekkers. Estimating and simulating with a random utility random opportunity model of job choice: Presentation and application to Belgium. *International Journal of Microsimulation*, 9(2):144–191, 2016.
- John Creedy and Nicolas Hérault. Decomposing inequality and social welfare changes: the use of alternative welfare metrics. Working Paper 8/11, Melbourne Institute of Applied Economic and Social Research, 2011.
- John K. Dagsvik and Zhiyang Jia. Labor supply as a choice among latent jobs: Unobserved heterogeneity and identification. *Journal of Applied Econometrics*, 31(3):487–506, 2016.
- John K. Dagsvik and Steinar Strøm. Sectoral labour supply, choice restrictions and functional form. *Journal of Applied Econometrics*, 21(6):803–826, 2006.
- André Decoster and Peter Haan. Empirical welfare analysis with preference heterogeneity. *International Tax and Public Finance*, 22(2):224–251, 2015.
- Marc Fleurbaey and François Maniquet. Fair income tax. *The Review of Economic Studies*, 73(1):55–83, 2006.
- Marc Fleurbaey and François Maniquet. Fairness and well-being measurement. *Mathematical Social Sciences*, 90:119–126, 2017. doi: 10.1016/j.mathsocsci.2017.02.005.
- Marc Fleurbaey and François Maniquet. Optimal income taxation theory and principles of fairness. *Journal of Economic Literature*, 56(3):1029–1079, 2018. doi: 10.1257/jel.20171238.
- Laurence Jacquet, Zhiyang Jia, and Thor O. Thoresen. How much does responsibility matter in fairness measurement? Working Paper 12418, CESifo, 2026.
- Robin Jessen. Why has income inequality in Germany increased from 2002 to 2011? a behavioral microsimulation decomposition. *Review of Income and Wealth*, 65(3):540–560, 2019. doi: 10.1111/roiw.12397.
- Zhiyang Jia and Thor O. Thoresen. Welfare effects of tax policy change when there are choice restrictions on labour supply. Discussion Paper 959, Statistics Norway, Research Department, 2021.
- Jannek Mühlhan. The German job miracle and its impact on income inequality: a decomposition study. *International Journal of Microsimulation*, 16(1):28–64, 2023. doi: 10.34196/ijm.00274.
- Guillermo Owen. Values of games with a priori unions. In Rudolf Henn and Otto Moeschlin, editors, *Mathematical Economics and Game Theory*, pages 76–88. Springer, Berlin, 1977. doi: 10.1007/978-3-642-45494-3_7.

- Mercedes Sastre and Alain Trannoy. Shapley inequality decomposition by factor components: some methodological issues. *Journal of Economics*, 77(S1):51–89, 2002. doi: 10.1007/BF03052500.
- Anthony F. Shorrocks. Inequality decomposition by factor components. *Econometrica*, 50(1):193–211, 1982. doi: 10.2307/1912537.
- Anthony F. Shorrocks. Decomposition procedures for distributional analysis: A unified framework based on the shapley value. *Journal of Economic Inequality*, 11:99–126, 2013.
- Holly Sutherland and Francesco Figari. Euromod: The european union tax-benefit microsimulation model. EUROMOD Working Paper EM8/13, Institute for Social and Economic Research, University of Essex, 2013.